AerialToken: Geometry-Guided Domain Generalization for Remote Sensing Semantic Segmentation

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***Abstract*—Domain-generalized semantic segmentation (DGSS) is critical for remote sensing and Earth observation imagery, where models trained on annotated satellite or unmanned aerial vehicle (UAV) imagery from a source region often suffer substantial performance degradation when deployed to unseen sensors, spatial resolutions, seasonal conditions, and geographic areas. However, acquiring dense pixel-level annotations for every new remote sensing scene remains costly and difficult to scale. Recent adapters built upon frozen visual foundation models (VFMs) have improved source-only generalization, but they still rely primarily on RGB appearance, whose spectral and textural statistics vary considerably across remote sensing domains. Although monocular depth provides structural cues, dense depth predictions in overhead imagery are sensitive to flight altitude, ground sampling distance, sensor intrinsics, and scene layout. We propose AerialToken, a geometry-aware framework that uses a frozen depth foundation model as an intermediate geometry token generator rather than a final depth prior. Predecoder representations from Depth Anything V2 are placed in parameter-free block-wise correspondence with a frozen visual encoder through geometry representation alignment, injected through layerwise dual route token calibration, and aggregated into geometry-guided decoder queries for the query-conditioned structural decoder. Experiments on two single-source cross-domain remote sensing protocols spanning satellite and UAV benchmarks show consistent improvements on unseen datasets without target-scene annotations. The results indicate that intermediate geometry representations provide more stable height-like structural cues than final dense depth maps where the domain shift alters depth scale, and that the two representations become comparable where it does not. The code is available at https://github.com/SY-Ch/AerialToken**

***Index Terms*—Domain generalization, intermediate geometry features, remote sensing semantic segmentation.**

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I. Introduction

DOMAIN-GENERALIZED semantic segmentation (DGSS) [1], [2], [3], [4] aims to learn pixel-level scene understanding from a labeled source domain and generalize it to unseen target domains without access to target annotations or test-time adaptation. This setting is particularly relevant to unmanned aerial vehicle (UAV) and remote sensing imagery [5], [6], where acquiring dense pixel-level labels is labor-intensive, costly, and difficult to repeat for every newly observed scene or geographic region. In real-world aerial mapping, environmental monitoring, and urban analysis, deployment data may differ substantially from the source domain due to changes in geographic regions, sensors, flight altitudes, imaging resolutions, seasons, illumination conditions, and land-cover compositions. From a photogrammetric and remote sensing perspective, improving such source-only generalization is therefore central to reducing repeated annotation costs and supporting scalable aerial scene interpretation.

The scientific difficulty of cross-domain transfer in aerial segmentation can be traced to an imaging-geometry asymmetry between unstable appearance statistics and relatively stable scene structure. Under overhead or near-overhead imaging, objects with distinct elevations are projected onto a 2-D image plane, causing different land-cover categories to exhibit similar co-planar appearances. For example, building roofs, road surfaces, and tree crowns may exhibit similar textures and tones in RGB, even though their vertical positions in the 3-D scene are entirely different. Appearance-level domain shift, including changes in illumination, sensor response, and seasonal material variation, further exacerbates this textural confusion but does not alter the vertical ordering among land-cover types. Roads and water remain near the terrain surface across cities and seasons, buildings consistently form locally elevated structures, and low vegetation occupies a narrow height band that is stable across datasets [7], [8], [9], [10]. This observation motivates a representation strategy that exploits

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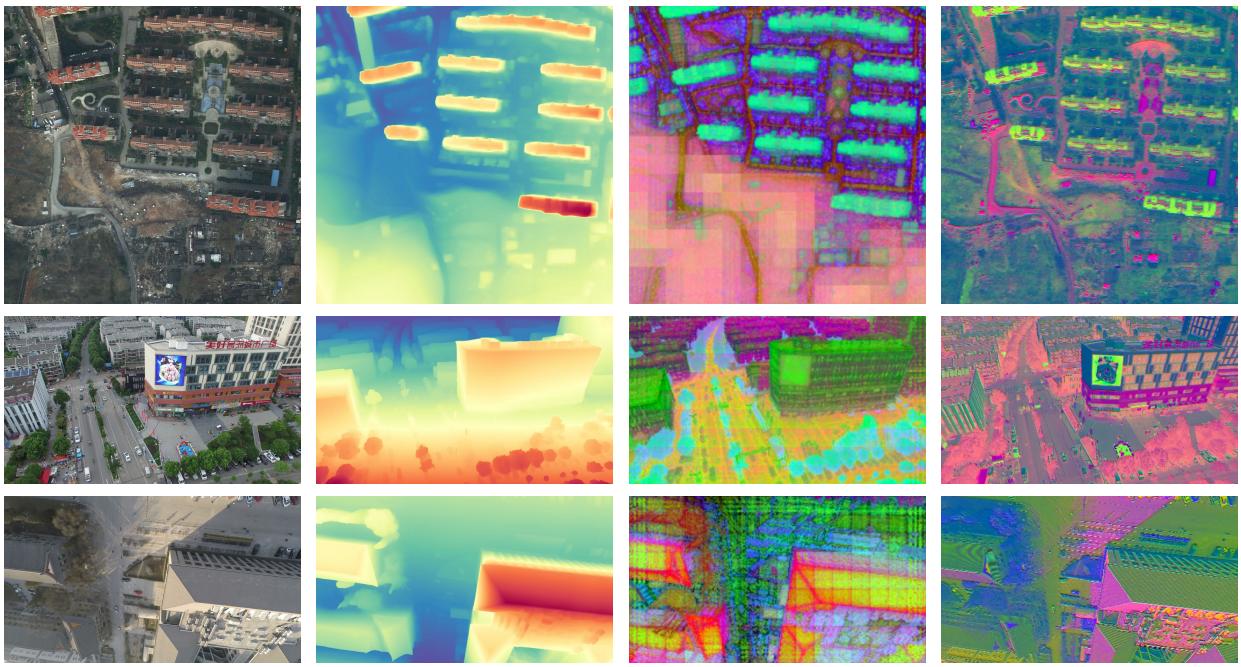


Fig. 1. Feature visualization on representative satellite and UAV scenes. (From left to right) We show the input image, monocular depth, PCA visualization of frozen DINOv2 features, and PCA visualization of our features. The frozen DINOv2 features are not well clustered on aerial imagery, especially in rear-view UAV scenes where feature collapse across different objects is more evident. In contrast, our geometry-enhanced features exhibit improved compactness and clearer structural separation, indicating better discrimination for major semantic regions such as buildings and roads.

height-related structural regularities of the physical scene as a more stable organizing cue than dataset-specific spectral or textural patterns.

Visual foundation model (VFM)-based DGSS methods reduce overfitting by keeping large pretrained encoders frozen and adapting only lightweight modules. However, their dominant signal still comes from RGB appearance [3], [4], making them vulnerable when appearance statistics vary substantially across domains. This limitation is particularly severe in aerial imagery, where viewpoint and scale changes further weaken the transferability of purely appearance-driven cues. As illustrated in Fig. 1, frozen DINOv2 features on representative satellite and UAV scenes exhibit weak compactness and poor structural separation, whereas geometry-enhanced features produce clearer semantic organization. Thus, the key is not merely to add a geometric modality, but to formulate geometry in a form that is compatible with overhead-image semantics and frozen VFM representations.

In perspective imagery, monocular depth is closely tied to camera-centric distance and often correlates with foreground-background separation [11], [12]. For overhead imagery, however, monocular depth is more naturally interpreted as a height-like field that encodes relative elevation, vertical stratification, and spatial layout. This distinction makes the choice of representation critical. Directly transferring the final dense prediction can provide useful cues in some target domains, but it also introduces several concrete failure modes when used as a fixed prior. First, overhead images compress elevated objects onto the ground plane, so roofs, tree crowns, roads, and shadows can have ambiguous height-image relationships when only RGB evidence is available. Second, changes in flight altitude and ground sampling distance alter object size and apparent relief, making the numeric scale of a dense monocular map inconsistent across UAV and satellite datasets. Third, even strong depth foundation models such as Depth Anything V2 [13] can exhibit scale drift across sensors and scene layouts because their final decoder is optimized for relative monocular depth rather than metrically calibrated aerial height. Therefore, aerial DGSS should exploit monocular depth as complementary geometric evidence while avoiding direct dependence on the final dense map as a fixed cross-domain prior.

Guided by this analysis, AerialToken is not designed as an incremental depth-fusion strategy, but as a geometry-token adaptation framework for aerial DGSS. It treats the frozen depth foundation model as a structural token generator, extracts predecoder geometry tokens, and injects them into frozen visual foundation representations through reliability-calibrated token regulation rather than final depth-map concatenation. First, geometry representation alignment constructs the geometry representation from the predecoder tokens of a frozen monocular geometry backbone and places it in one-to-one correspondence with the visual tokens at every encoder block. Because the two frozen backbones are architecturally homogeneous, this correspondence needs no resampling, no channel projection, and no trainable parameters, so it is a property of how the geometry representation is constructed rather than a separate module inserted into the network. Second, layerwise dual route token calibration balances appearance-driven and geometry-driven token affinity, suppresses unreliable geometry responses, and writes geometric evidence back into the visual encoder through

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bounded residual updates. Finally, the query-conditioned structural decoder transforms calibrated layer-specific tokens into geometry-guided decoder queries, refining cross-scale context, elongated structures, and object boundaries for aerial mask prediction.

We evaluate AerialToken on two representative single-source cross-domain protocols with unified remapping and mean intersection-over-union (mIoU) computation. The first trains on VDD [7] and tests on VDD, UDD [14], and UAVid [15], targeting within-UAV transfer. The second trains on LoveDA [8] and tests on LoveDA, FLAIR [16], FloodNet [17], Potsdam [9], and Vaihingen [10], covering broader cross-dataset remote sensing transfer. Across both settings, injecting intermediate geometry cues into a frozen VFM consistently improves performance on unseen domains.

Our main contributions are as follows.

1) We formulate geometry use in remote sensing DGSS from the perspective of overhead-imaging failure modes, identifying height ambiguity, ground-sampling-distance variation, and monocular depth scale drift as key reasons to avoid relying only on direct dense-depth transfer.
2) We propose AerialToken, a geometry-aware remote sensing DGSS framework that pairs intermediate geometry representations with every block of a frozen visual encoder without added parameters, calibrates appearance and geometry cues through bounded dual-route token interaction, and propagates the resulting geometry-enhanced semantics into the query-conditioned structural decoder.
3) We validate AerialToken on two representative single-source cross-domain remote sensing segmentation protocols spanning satellite and UAV benchmarks, with deterministic label remapping, controlled component ablations, and diagnostic analyses demonstrating consistent improvements on unseen domains.

II. Related Work

A. VFM-Based Domain Generalization

DGSS has gradually moved from explicitly reducing domain-sensitive statistics to adapting stronger pretrained representations. Early studies improved robustness by weakening style-dependent feature bias [18] or strengthening structured prediction during decoding [19]. With the emergence of vision foundation models, parameter-efficient adaptation has become more common. Representative studies such as MRFP [20] and REIN [4] show that a frozen backbone with a small number of trainable modules can transfer broad visual priors to unseen domains more effectively than conventional full fine-tuning. This trend has also extended to remote sensing [21], [22]. VFM-based overhead parsing [23] has shown clear cross-domain potential [24]. Auxiliary priors from depth-guided tuning [25], depth-language collaboration [6], and geospatial foundation modeling [26] further improve robustness under larger distribution gaps [27]. Even so, most existing methods still enhance transferability mainly through visual representation tuning [28]. Structural cues that are particularly important in aerial scenes remain underused.

AerialToken differs from existing depth- or geometry-assisted VFM adapters in the source, role, and injection mechanism of the geometric signal. DepthForge [25] injects depth-aware learnable tokens into frozen VFM layers and refines visual features through depth-guided decoding, while other depth-anything-based adapters commonly use depth as an auxiliary prediction, prompt, attention cue, or decoder-side refinement signal. Vireo [6] couples depth with language-level priors for broader semantic guidance, and CrossEarth [26] improves remote sensing robustness through geospatial foundation pretraining, Earth-style injection, and multitask training. In contrast, AerialToken introduces no new foundation pretraining, uses no target-domain data, and keeps both the visual and geometry backbones frozen. More importantly, it discards the final dense depth prediction and treats the frozen depth foundation model as a predecoder structural token generator. The resulting intermediate geometry tokens are aligned to selected visual layers, used as reliability-gated regulators of visual token affinity through bounded dual-route calibration, and then aggregated into geometry-guided queries for the query-conditioned structural decoder. Thus, geometry in AerialToken shapes both frozen VFM adaptation and mask-query formation, rather than serving only as an auxiliary dense map, a decoder-side cue, or an unconstrained fused modality.

B. Geometry in Domain-Generalized Segmentation

Geometry has long been regarded as a useful complement to appearance in overhead imagery. Studies based on distance-map supervision [29] or joint semantic-height learning [30] show that geometric information helps distinguish categories with similar textures but different vertical organization. When additional modalities are available, multilevel and cross-branch fusion strategies [31], [32] further improve category separation by integrating image features with height-related cues across semantic scales [33]. More recent work suggests that generic segmentation foundation models transferred from natural images [34], [35] do not fully fit overhead scenes. Moreover, large monocular depth models such as Depth Anything V2 [13] make geometry much easier to obtain in practice. Dedicated aerial height-estimation networks [36] also strengthen structural modeling for overhead imagery. Final dense depth or height values can be helpful in some scenes, but they remain sensitive to flight altitude, sensor settings, and scene composition. This limits their direct use as standalone fixed priors for aerial DGSS. Therefore, our method uses intermediate geometry features, which are more stable across domains and better aligned with aerial semantics.

C. UAV and Remote Sensing Semantic Segmentation

Compared with ground-view parsing, UAV and remote sensing semantic segmentation faces more severe scale variation, denser small objects, more cluttered spatial layouts, and stronger appearance differences across platforms and regions [37]. These challenges are reflected in representative benchmarks for UAV urban scenes, urban-rural transfer, and broader cross-region land-cover parsing. Typical examples

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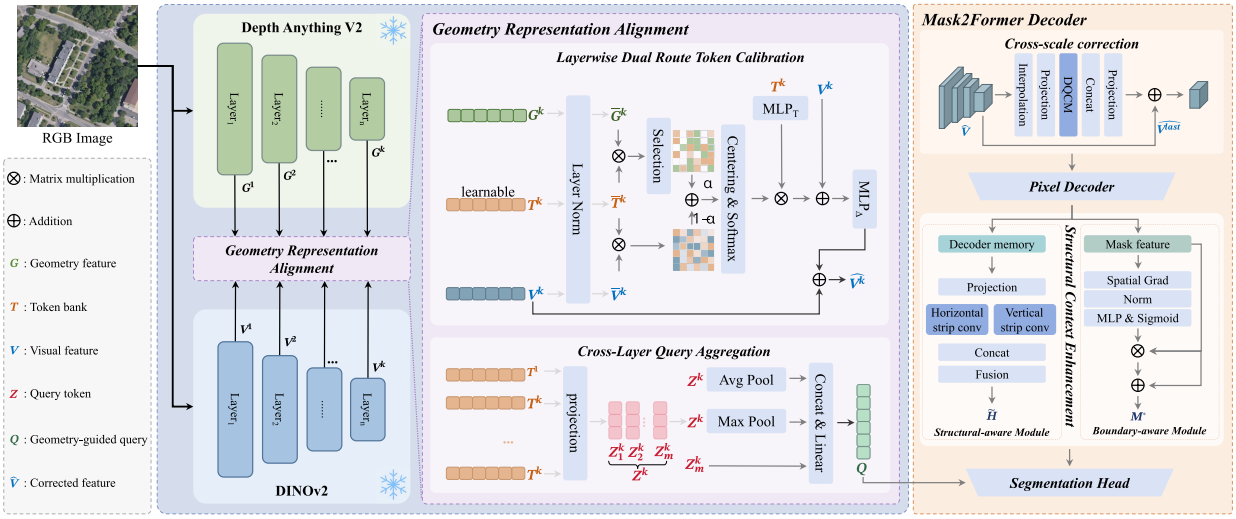


Fig. 2. Overall framework of AerialToken, which pairs intermediate geometry representations block by block with a frozen visual backbone through geometry representation alignment, calibrates visual and geometry cues through layerwise dual route token calibration, aggregates cross-layer token banks into geometry-guided decoder queries, and performs geometry-guided mask decoding for aerial DGSS.

include UAVid [15], LoveDA [8], FLAIR [16], and cross-city benchmarks [38]. To address these issues, recent studies improve geospatial parsing through foreground-aware relation modeling [39] and more efficient context aggregation for very-high-resolution imagery [40], [41], [42], [43]. Structure-sensitive categories have received particular attention. For road extraction, semantic–spatial feature refinement [44] and context perception combined with cross-spatial-scale interaction [45] both show that elongated, topologically connected structures benefit from explicitly separating semantic evidence from spatial detail. This is consistent with our own observation that roads are among the categories most improved by geometry-aware calibration. Comparable structure-aware designs appear beyond optical land-cover parsing, e.g., in spatial–frequency feature learning for infrared small target detection [46], and in referring remote sensing image segmentation, where intra- and cross-scale feature interaction aligns linguistic queries with multiscale visual evidence [47]. These methods enrich the feature interaction pattern within a single supervised domain, whereas our objective is to keep a frozen representation transferable to domains that are never seen during training.

A complementary line of work reduces annotation cost by weakening the form of supervision rather than by transferring across domains. Recent examples model label distributions to obtain better pseudomasks [48], and exploit segmentation foundation models for weakly supervised remote sensing segmentation through adaptive token linking [49] or SAM-guided cross-teaching [50]. These approaches still assume that some annotation, however coarse, is available in the deployment scene. Domain generalization assumes no target annotation at all, so the two directions attack the annotation bottleneck from opposite ends and are largely orthogonal; combining weak target supervision with a geometry-calibrated frozen backbone is an interesting direction that we leave to future work.

Nevertheless, most of these advances are still developed for supervised or in-domain remote sensing segmentation. As a result, the cues they learn remain largely appearance-centered when transferred to unseen aerial domains. Our method targets this setting by introducing geometry-aware representations that provide more transferable structural evidence for aerial cross-domain segmentation.

III. Method

A. Overview

As shown in Fig. 2, we propose AerialToken, a geometry-aware remote sensing semantic segmentation framework that enhances a frozen visual backbone with intermediate geometry representations and decodes the resulting features with query-conditioned structural reasoning. Given an input overhead image x from satellite or UAV platforms, a frozen visual backbone $\mathcal{B}_v$ produces selected token sequences $\{V^k\}_{k\in\mathcal{K}}$ and multiscale feature maps $\{F_i\}_{i=1}^{S}$, while a frozen geometry backbone $\mathcal{B}_g$ produces intermediate geometry tokens $\{G^k\}_{k\in\mathcal{K}}$. The geometry tokens are used in two ways. First, they calibrate the visual tokens through a layerwise dual-route token bank. Second, the token banks are aggregated across layers to form geometry-guided decoder queries. The calibrated multiscale features and the query set are then fed into a Mask2Former-style decoder to predict semantic masks.

B. Geometry Representation Alignment

A key observation behind AerialToken is that, in remote sensing imagery acquired from satellite or UAV platforms, the absolute scale of monocular depth predictions is strongly affected by flight altitude, ground sampling distance, and camera intrinsics. Therefore, the final dense depth values are not sufficiently stable to serve as the only cross-domain semantic cue. Instead, we draw geometry information from the intermediate representations of the frozen geometry backbone.

Given an input image x, the visual backbone and the geometry backbone produce a set of selected token sequences and intermediate geometry representations

$$\{V^k, G^k\}_{k\in\mathcal{K}}, \qquad V^k \in \mathbb{R}^{N_k\times C}, \quad G^k \in \mathbb{R}^{N_k\times C} \tag{1}$$

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where $\mathcal{K}$ denotes the selected encoder layers, N_k is the token length at layer k, and C is the embedding dimension.

Here, G^k denotes the intermediate geometry token read from the frozen depth backbone *before* its depth decoder. Depth Anything V2 is built on a DINOv2 encoder fine-tuned for monocular depth estimation, so at the same vision Transformer (ViT) scale its block structure is identical to that of the DINOv2 visual backbone. We exploit this homogeneity deliberately. Both branches use a DINOv2 ViT-L encoder whose patch embedding is resampled to a 16×16 patch size, so that a 512×512 input divides evenly ($512/16 = 32$). The two encoders therefore produce token grids of identical size (32×32, i.e., $N_k = 1024$), identical embedding dimension ($C = 1024$), and an identical number of blocks. Consequently $\mathcal{K} = \{0, 1, \ldots, 23\}$ indexes *all* encoder blocks, and the geometry token G^k is paired one-to-one with the visual token V^k at every block. The resampling that makes this correspondence possible is applied once, offline, to both encoders. DINOv2 ViT-L and the Depth Anything V2 encoder are both released with a 14×14 patch-embedding kernel and a 37×37 position-embedding grid, matching their 518×518 pretraining resolution. For each branch, we resize the patch-embedding kernel to 16×16 and the position-embedding grid to 32×32, both by bicubic interpolation, and copy the class-token embedding unchanged; nothing else in either encoder is modified. Because $512/16 = 32$, the stored grid is exactly the grid a 512×512 input produces, so no positional embedding is interpolated at run time. The released conversion script performs both steps and writes the two resampled encoders into a single file, and running it on the two public checkpoints regenerates precisely the frozen backbone used for every result reported here. No spatial interpolation, channel projection, or additional geometry pretraining stage is needed to align them; the only operation applied before use is the parameter-free layer normalization defined in Section III-C. This is a design choice rather than an implementation shortcut: a geometry encoder of a different scale would reintroduce resolution mapping, channel projection, and block-correspondence heuristics that the present formulation avoids by construction. Geometry representation alignment therefore names this parameter-free block-wise correspondence and the construction of the geometry representation itself, not a learned module that could be inserted or removed on its own. It adds no trainable parameters and performs no resampling, and its content is the decision to build the geometry representation from predecoder tokens and to pair it with the visual tokens at every block. Its contribution is accordingly measured by replacing that representation with the final dense depth map while holding the rest of the network fixed, as reported in Section IV.

C. Layerwise Dual Route Token Calibration

To inject geometry information into the frozen visual encoder, we assign an independent learnable token bank to each selected layer

$$T^k \in \mathbb{R}^{M \times C}, \quad k \in \mathcal{K} \tag{2}$$

where M is the token bank size. These tokens are layer-specific and do not propagate across layers, which keeps the adaptation lightweight and stable. For parameter efficiency, each token bank is stored in low-rank factorized form $T^k = A^k B^k$ with $A^k \in \mathbb{R}^{M \times 16}$ and $B^k \in \mathbb{R}^{16 \times C}$, so that the trainable token parameters scale with $16(M + C)$ rather than MC. The factorization rank is fixed to 16 in all experiments and is never varied; M is therefore the only quantity that controls token-bank capacity.

For clarity, we omit the batch dimension in the following formulation. We first normalize the visual features, geometry features, and token bank

$$\bar{V}^k = \mathrm{LN}\left(V^k\right), \quad \bar{G}^k = \mathrm{LN}\left(G^k\right), \quad \bar{T}^k = \mathrm{LN}\left(T^k\right). \tag{3}$$

1) Dual-Route Token Affinity: We compute token affinities from both the visual and geometry branches

$$S_v^k = \frac{\bar{V}^k \left(\bar{T}^k\right)^\top}{\sqrt{C}\,\tau_v}, \quad S_g^k = \frac{\bar{G}^k \left(\bar{T}^k\right)^\top}{\sqrt{C}\,\tau_g} \tag{4}$$

where $S_v^k, S_g^k \in \mathbb{R}^{N_k \times M}$. The temperatures are parameterized as $\tau_v = \mathrm{softplus}(\theta_v) + \epsilon_\tau$ and $\tau_g = \mathrm{softplus}(\theta_g) + \epsilon_\tau$, where θ_v and θ_g are unconstrained learnable scalars, and $\epsilon_\tau > 0$ is a small constant. This guarantees positive temperatures and prevents the similarity order from being reversed.

Since not every geometry response is equally reliable under aerial domain shift, we further suppress low-energy geometry positions. Specifically, we compute a geometry energy score for each token

$$e_i^k = \frac{1}{C} \sum_{c=1}^{C} \left| \bar{G}_{i,c}^k \right| \tag{5}$$

and retain only the top-ρ fraction of positions

$$m_i^k = \mathrm{sg}\left(\mathbf{1}\left[e_i^k \in \mathrm{TopK}_\rho\left(\{e_j^k\}_{j=1}^{N_k}\right)\right]\right). \tag{6}$$

Here, $\mathrm{sg}(\cdot)$ denotes stop-gradient. The hard top-ρ selection is recomputed at each forward pass but is not differentiated through; gradients still flow through the selected geometry logits S_g^k and the trainable token bank.

Three properties of this screening step are worth making explicit. First, it is applied *identically* during training and inference: m^k is recomputed in every forward pass from the current geometry energy, and no branch of the model disables it at test time. The training and inference pipelines therefore share exactly the same screening operator, and the diagnostic row that disables top-ρ in Section IV is obtained precisely by switching this operator off at inference. Second, the geometry backbone $\mathcal{B}_g$ is frozen throughout training, so no gradient is ever propagated into it; the screening cannot under-train the geometry branch, because the branch has no trainable parameters to begin with. The mask only determines which spatial positions contribute geometry-driven affinity to the trainable token bank. Third, positions outside the retained set are not removed from the computation: setting $m_i^k = 0$ suppresses only the geometry term at position i, while the visual term $(1-\alpha)S_v^k$ is preserved in full, so every spatial position still receives a calibrated residual update. Top-ρ is thus a selector over *which cue* is trusted at each position, not a sparsifier over which positions are processed. Let $m^k = [m_1^k, \ldots, m_{N_k}^k]^\top \in \{0, 1\}^{N_k}$

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and let $\mathbf{1}_M$ be an all-ones vector of length M. The final token affinity is then obtained by adaptively mixing the two branches

$$A^k = \mathrm{Softmax}_M\left(\mathrm{ColCtr}\left((1-\alpha)\,S_v^k + \alpha\left(\left(m^k\mathbf{1}_M^\top\right)\odot S_g^k\right)\right)\right) \tag{7}$$

where $\alpha = \sigma(\eta_\alpha)$ is a learnable mixing coefficient, $\odot$ denotes element-wise multiplication, and $\mathrm{Softmax}_M$ is applied along the token-bank dimension. For $X \in \mathbb{R}^{N_k\times M}$, $\mathrm{ColCtr}(X) = X - \mathbf{1}_{N_k}\mu_X^\top$ with $\mu_X = 1/N_k X^\top \mathbf{1}_{N_k}$ denotes column-wise centering across spatial positions. This notation avoids conflict with the embedding dimension C. Because column-wise centering subtracts a different offset for each token-bank column rather than a row-wise constant, it changes the row-wise softmax distribution and suppresses globally dominant token responses. This design balances appearance-driven and geometry-driven matching without allowing the geometry branch to overwrite visual evidence directly.

2) Residual Token Injection: The fused token affinity is converted into a feature residual and written back to the visual encoder

$$\Delta^k = \mathrm{MLP}_\Delta\left(A^k\,\mathrm{MLP}_T\left(T^k\right) + V^k\right). \tag{8}$$

To prevent overly strong perturbations, we apply relative clipping

$$\hat{\Delta}^k = \mathrm{ClipRel}\left(\Delta^k, V^k; r\right) \tag{9}$$

with

$$\mathrm{ClipRel}\,(d, v; r) = d\cdot\min\left(1, \frac{r\|v\|_F}{\|d\|_F + \epsilon}\right). \tag{10}$$

The calibrated layer output is

$$\hat{V}^k = V^k + s\,\hat{\Delta}^k \tag{11}$$

where s is a learnable residual scale. This bounded update injects geometry information through a controlled residual path, which is important for keeping the frozen visual representation stable under aerial domain shift.

3) Cross-Layer Query Aggregation: In addition to encoder calibration, the token banks are also used to construct decoder queries. Each layer-specific token bank is first projected into the query space

$$Z^k = T^k W_q, \quad Z^k \in \mathbb{R}^{M\times d} \tag{12}$$

where d is the decoder query dimension and $W_q \in \mathbb{R}^{C\times d}$. Let $\mathbf{z}_m^k$ be the mth token in Z^k, and let $k_{\mathrm{last}} = \max(\mathcal{K})$ be the last selected layer. We then aggregate the tokens across layers by combining max pooling, average pooling, and the final-layer token

$$\mathbf{q}_m = W_{\mathrm{agg}}\left[\max_{k\in\mathcal{K}}\mathbf{z}_m^k\,;\,\mathrm{avg}_{k\in\mathcal{K}}\mathbf{z}_m^k\,;\,\mathbf{z}_m^{k_{\mathrm{last}}}\right]$$
$$m = 1,\ldots,M \tag{13}$$

where $[\cdot;\cdot;\cdot]$ denotes concatenation, and W_{agg} is a shared aggregation projection. The resulting query set is

$$Q = \begin{bmatrix}\mathbf{q}_1^\top\\ \vdots\\ \mathbf{q}_M^\top\end{bmatrix} \in \mathbb{R}^{M\times d}. \tag{14}$$

This query set serves as the geometry-guided query embedding for decoding.

D. Query-Conditioned Structural Decoder

The geometry-enhanced multiscale backbone features are passed to a Mask2Former-style segmentation head. Let $\{F_i\}_{i=1}^S$ be the multiscale feature maps from the calibrated visual encoder, where F_1 is the highest-resolution feature. Before Transformer decoding, we apply a lightweight cross-scale correction to improve structural sensitivity in aerial scenes

$$\hat{F}_1 = F_1 + \Phi\left(\{\Pi_i(F_i)\}_{i=1}^S\right) \tag{15}$$

where $\Pi_i(\cdot)$ denotes bilinear resizing to the spatial resolution of F_1, and $\Phi(\cdot)$ denotes gated cross-scale fusion. The refined multiscale input is

$$\hat{\mathcal{F}} = \left\{\hat{F}_1, F_2, \ldots, F_S\right\}. \tag{16}$$

A pixel decoder first transforms $\hat{\mathcal{F}}$ into a set of multiscale memory features and a high-resolution pixel embedding map $P \in \mathbb{R}^{HW\times d}$. The query set from Section III-C3 initializes the Transformer decoder

$$Q^0 = Q. \tag{17}$$

At decoder layer t, the queries interact with the multiscale memory through self-attention and cross-attention

$$Q^{t+1} = \mathrm{Dec}^t\left(Q^t, \mathcal{M}_{\sigma(t)}\right) \tag{18}$$

where $\mathcal{M}_{\sigma(t)}$ denotes the selected memory level at layer t, and $\mathrm{Dec}^t(\cdot)$ denotes the standard Mask2Former decoding block. Because the initial queries are aggregated from cross-layer geometry-calibrated token banks, the decoding process is guided toward regions that remain structurally consistent across aerial domains.

For each final query $\mathbf{q}_m^{(T)}$ after the last decoder layer, the classification score is predicted as

$$\mathbf{p}_m = \mathrm{Linear}_{\mathrm{cls}}\left(\mathbf{q}_m^{(T)}\right) \tag{19}$$

and the mask embedding is computed as

$$\mathbf{w}_m = \mathrm{MLP}_{\mathrm{mask}}\left(\mathbf{q}_m^{(T)}\right). \tag{20}$$

The corresponding mask logit is obtained by projection onto the pixel embedding map

$$\hat{Y}_m(h,w) = P_{h,w}^\top \mathbf{w}_m \tag{21}$$

where $P_{h,w} \in \mathbb{R}^d$ denotes the pixel embedding at location (h, w). The final semantic prediction is produced by combining the classification scores and mask logits in the standard set-prediction manner.

1) Relation to the Vanilla Mask2Former Decoder: Three components differ from a standard Mask2Former head, and the remainder is deliberately left unchanged.

1) *Query Formation:* A vanilla head initializes the query positional embedding and the query feature as free parameters that belong to the decoder alone. In our decoder, both are derived from Q: the query embedding is Q itself, and the query feature is obtained by a single linear map of Q. The difference is not that the queries become input-dependent. By (12)–(14), Q is a function of the token banks only, so it is fixed at inference exactly as a vanilla codebook is. The difference is that

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the queries are not an independent codebook: the same layer-specific banks T^k that inject geometry into the encoder also generate them, so query formation and encoder calibration share parameters and the geometry route shapes the query set throughout training.

2) *Cross-Scale Correction:* The gated fusion $\Phi(\cdot)$ writes a residual correction into the last two of the three multiscale memory levels and into the mask-embedding map, using a strip-shaped context kernel suited to the elongated structures common in overhead imagery.
3) *Boundary-Aware Refinement:* The mask-embedding map is modulated by a gate computed from a fixed, non-learned gradient operator, so boundary evidence enters as a bounded multiplicative residual instead of an additional supervised prediction branch. The pixel decoder, the nine Transformer decoder layers, the masked-attention mechanism, the Hungarian matching, and the classification, mask, and dice losses are inherited from Mask2Former without modification. The distinguishing property of the proposed decoder is therefore not extra capacity but the fact that its queries carry cross-layer geometric evidence into mask formation; components (2) and (3) are bounded residual refinements whose scales are initialized near zero, and their individual contributions are isolated in Section IV.

IV. Experiments

A. Benchmarks and Evaluation Protocols

We design two single-source cross-domain protocols to evaluate AerialToken under different domain-shift intensities. The first is the VDD-source protocol, which trains on VDD and evaluates on VDD, UDD, and UAVid to measure transfer among UAV-view domains that share similar viewpoints but differ in city, sensor, and annotation style. The second is the LoveDA-source protocol, which trains on LoveDA and evaluates on LoveDA, FLAIR, FloodNet, Potsdam, and Vaihingen, thereby introducing broader sensor diversity, geographic variation, and annotation heterogeneity. Together, these protocols cover both moderate and strong forms of domain shift in aerial segmentation. This revision extends each protocol with additional unseen targets that are used for evaluation only and never for training or model selection: OpenEarthMap [54] is added to the LoveDA-source protocol, and SkyScapes-Dense [55] and AeroScapes [56] are added to the VDD-source protocol. The main cross-domain comparison therefore contains five unseen targets under the LoveDA-source protocol, while the VDD-source summary reports the source domain together with four unseen domains. The criteria by which these benchmarks were selected are given later in this section, and their label mappings are listed in Section IV-F.

We report mIoU as the primary metric after deterministic class-id conversion, and use per-class intersection over union (IoU) for fine-grained analysis. The evaluation follows the standard DGSS protocol: target-domain images are used only at test time, with no target labels during training and no test-time adaptation.

To ensure fair comparison under the unified label spaces defined for each protocol, all competing methods were retrained and re-evaluated under our experimental settings rather than directly copied from the original papers. In our reimplementation, REIN and CrossEarth are treated as DINOv2 ViT-L frozen-VFM baselines under the shared backbone setting, making them representative VFM-style baselines under the unified protocol. The protocol shared by every compared method covers the data and the entire evaluation path: a batch size of 4, a 512×512 training crop, random resize, crop and flip augmentation, polynomial learning-rate decay with power 0.9, sliding-window inference with a 512×512 window and a stride of 341, the same dataset remapping rules, the same target-domain masks, and the same mIoU computation pipeline. Optimization is matched within each model family rather than forced across families. REIN, CrossEarth, and AerialToken adapt the same frozen DINOv2 ViT-L backbone and share the AdamW configuration described below. The remaining compared methods are recent remote sensing segmentation architectures trained end-to-end, and for these we retain the optimizer published by their authors, a Lookahead-wrapped AdamW, because it is tuned to the architecture rather than to the protocol, and overriding it would handicap the baseline instead of equalizing the comparison. Every method is trained for 40 000 iterations, and in every case we report the final checkpoint of that schedule, so no target label enters model selection. Baseline hyperparameters were initialized from the official implementations and were not selectively tuned on target-domain labels. The REIN visual-only row in Table I provides a matched DINOv2 ViT-L baseline under the same training and evaluation protocol, while the contributions of the geometry route and of the decoder modifications are isolated separately in the ablation studies.

TABLE I
Component Ablation Study Under the LoveDA-Source Protocol (%)

Configuration	FLAIR	FloodNet	Pots.	Vaih.	Mean
REIN	43.85	38.39	27.05	32.51	35.45
+ depth map (early fusion)	43.42	37.62	27.08	32.57	35.17
+ geometry-token route	44.66	45.06	29.34	36.77	38.96
+ intermediate geometry	43.51	44.55	29.45	41.56	39.77
+ dual-route calib.	43.85	44.42	29.49	40.35	39.53
+ structural decoder	41.99	45.21	28.98	38.20	38.60
AerialToken	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**

B. Implementation Details

Our implementation is built upon the MMSegmentation framework [57]. We optimize the model with AdamW, using an initial learning rate of 1×10^{-4} under the LoveDA-source protocol and 6×10^{-5} under the VDD-source protocol, a weight decay of 0.05, an epsilon of 1×10^{-8}, and β coefficients of $(0.9, 0.999)$. The learning rate is scheduled by a polynomial decay policy over 40 000 iterations, with the power set to 0.9 and the minimum learning rate fixed at 0. All models use a batch size of 4 and a 512×512 training crop, and inference uses a sliding window of 512×512 with a stride of 341. REIN and CrossEarth adapt the same frozen backbone and are trained with the same optimizer settings as above;

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TABLE II

SUMMARY MIOU COMPARISON UNDER THE LOVEDA-SOURCE PROTOCOL (%). METHODS ARE ORDERED BY MEAN MIOU. BEST RESULTS ARE SHOWN IN BOLD

Method	Venue	FLAIR	FloodNet	Potsdam	Vaihingen	OpenEarthMap	Mean
LSKNet-S [51]	IJCV 2025	21.72	19.43	12.86	5.66	18.27	15.59
LSKNet-T [51]	IJCV 2025	22.41	27.21	14.47	6.75	20.36	18.24
SFANet [52]	RS 2024	31.97	16.30	17.63	19.68	32.61	23.64
D2LS [53]	ICCV 2025	33.20	20.37	19.88	21.51	32.22	25.44
REIN [4]	CVPR 2024	43.85	38.39	27.05	32.51	52.01	38.76
CrossEarth [26]	TPAMI 2025	43.56	40.71	27.77	35.07	52.75	39.97
Ours	-	**46.41**	**45.44**	**35.57**	**42.67**	**54.61**	**44.94**
Gain over best prior		+2.56	+4.73	+7.80	+7.60	+1.86	+4.97

TABLE III

CLASS-WISE IOU COMPARISON UNDER THE LOVEDA-SOURCE PROTOCOL (%)

Method	Building	Road	Water	Barren	Forest	Agricultural	mIoU
FLAIR							
LSKNet-S [51]	27.33	8.86	27.18	6.01	34.51	26.47	21.72
LSKNet-T [51]	26.63	8.51	28.90	7.19	31.85	**31.35**	22.41
SFANet [52]	62.10	17.20	34.28	10.33	53.46	14.47	31.97
D2LS [53]	59.42	17.14	48.57	**14.08**	47.72	12.24	33.20
REIN [4]	75.80	19.97	71.98	8.09	60.01	27.27	43.85
CrossEarth [26]	75.36	20.51	**73.38**	4.62	60.12	27.39	43.56
Ours	**77.03**	**35.23**	66.56	9.15	**61.40**	29.11	**46.41**
FloodNet							
LSKNet-S [51]	24.11	6.44	27.69	/	15.32	23.58	19.43
LSKNet-T [51]	23.37	4.72	31.07	/	23.36	**53.54**	27.21
SFANet [52]	22.69	6.81	23.07	/	28.89	0.02	16.30
D2LS [53]	31.54	4.53	25.00	/	40.79	0.00	20.37
REIN [4]	70.08	19.18	51.42	/	47.47	3.79	38.39
CrossEarth [26]	71.08	23.31	53.19	/	**54.50**	1.49	40.71
Ours	**71.96**	**43.06**	**56.39**	/	53.45	2.35	**45.44**
Potsdam							
LSKNet-S [51]	37.57	3.35	/	/	1.40	9.11	12.86
LSKNet-T [51]	36.12	2.85	/	/	0.82	**18.08**	14.47
SFANet [52]	50.59	17.27	/	/	1.89	0.77	17.63
D2LS [53]	58.89	18.83	/	/	1.18	0.64	19.88
REIN [4]	79.94	25.26	/	/	2.58	0.43	27.05
CrossEarth [26]	82.12	25.93	/	/	2.62	0.41	27.77
Ours	**84.30**	**52.91**	/	/	**4.46**	0.62	**35.57**
Vaihingen							
LSKNet-S [51]	11.10	11.54	/	/	0.00	0.00	5.66
LSKNet-T [51]	12.13	14.88	/	/	0.00	0.00	6.75
SFANet [52]	57.89	17.82	/	/	2.90	0.13	19.68
D2LS [53]	60.34	23.50	/	/	1.10	1.12	21.51
REIN [4]	82.55	32.85	/	/	11.00	3.65	32.51
CrossEarth [26]	86.44	34.93	/	/	**16.10**	2.81	35.07
Ours	**88.59**	**62.26**	/	/	14.81	**5.03**	**42.67**
OpenEarthMap							
LSKNet-S [51]	29.94	10.50	21.74	10.64	13.36	23.47	18.27
LSKNet-T [51]	34.91	9.00	23.14	9.20	15.75	30.14	20.36
SFANet [52]	63.30	27.86	28.24	**21.53**	30.43	24.29	32.61
D2LS [53]	59.94	28.00	46.69	18.17	23.22	17.30	32.22
REIN [4]	83.07	48.56	**68.85**	7.68	**46.21**	57.69	52.01
CrossEarth [26]	82.57	49.71	67.92	10.69	46.15	**59.45**	52.75
Ours	**83.41**	**65.92**	64.41	14.21	44.80	54.89	**54.61**

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TABLE IV
SUMMARY MIOU COMPARISON UNDER THE VDD-SOURCE PROTOCOL (%). METHODS ARE ORDERED BY MEAN MIOU. BEST RESULTS ARE SHOWN IN BOLD

Method	Venue	VDD	UDD	UAVid	SkyScapes	AeroScapes	Mean
LSKNet-T [51]	IJCV 2025	44.48	25.96	19.33	19.23	17.11	25.22
LSKNet-S [51]	IJCV 2025	44.37	26.45	19.91	20.67	16.24	25.53
D2LS [53]	ICCV 2025	70.41	49.50	46.69	30.71	27.22	44.91
SFANet [52]	RS 2024	76.78	53.43	49.39	34.45	23.65	47.54
REIN [4]	CVPR 2024	75.03	57.87	67.53	46.17	33.16	55.95
CrossEarth [26]	TPAMI 2025	80.47	62.19	70.22	47.30	28.52	57.74
Ours	-	**83.27**	**64.66**	**70.54**	**50.45**	**41.77**	**62.14**
Gain over best prior		+2.80	+2.47	+0.32	+3.15	+8.61	+4.40

the four end-to-end baselines keep the optimizer published by their authors, as described in Section IV-A. Every model reported in this article, including each ablation variant and each compared baseline, is trained once with the random seed fixed to 4307, and no reported result is selected from repeated trials. Quantifying variation across random seeds would require repeated training, and its absence is a limitation of this study.

For reference, the three frozen-VFM methods compared here learn almost the same number of parameters: 23.57 M for REIN and CrossEarth against 23.97 M for AerialToken, a difference of 1.7%, of which the token adapter accounts for 2.99 M in all three. The larger total for AerialToken, 663.32 M against 327.77 M, is entirely the second frozen encoder, which receives no gradients.

For reproducibility, we summarize the key hyperparameters of the proposed modules, all of which are fixed across both protocols and all datasets. The token bank size is $M = 100$ with factorization rank 16. The geometry screening ratio is $\rho = 0.25$, and the relative clipping bound is $r = 0.10$. The residual scale s is initialized to 1×10^{-3}, and the two affinity temperatures are initialized to $\tau_v = \tau_g = 0.7$. The mixing coefficient is parameterized as $\alpha = \sigma(\eta_\alpha)$ with η_α initialized to -1, i.e., $\alpha_0 \approx 0.27$; this starts training with the visual route dominant and allows the geometry weight to grow only if it reduces the training loss. The cross-scale correction is applied to the last two of the three multiscale memory levels. The multiscale features $\{F_i\}_{i=1}^{4}$ supplied to the pixel decoder are read from encoder blocks $\{7, 11, 15, 23\}$; note that these are the feature-pyramid taps for the segmentation head and are unrelated to the geometry injection, which operates at every block. Sensitivity to M, α, ρ, and r is reported in Section IV.

The RGB encoder is a frozen DINOv2 ViT-L backbone with a 16×16 patch embedding, and the geometry branch is a frozen Depth Anything V2 model whose encoder is the same DINOv2 ViT-L at the same patch size. The geometry branch supplies intermediate encoder features to every VFM layer before the depth decoder, where the layerwise dual route token calibration module injects these depth-related cues into the frozen visual backbone and aggregates the cross-layer token banks into decoder query embeddings. During training, only the token parameters, lightweight projection layers, decoder, and prediction heads are updated; both the DINOv2 visual encoder and the Depth Anything V2 geometry encoder remain frozen.

C. Performance Comparison

Table II reports the quantitative results under the LoveDA-source protocol using the unified remapping and mIoU pipeline. Our method achieves the best performance on all five target datasets, reaching 46.41 mIoU on FLAIR, 45.44 on FloodNet, 35.57 on Potsdam, 42.67 on Vaihingen, and 54.61 on OpenEarthMap, with the highest mean mIoU of 44.94. Compared with the strongest previous method, the proposed model improves the average performance by 4.97 points, including gains of 2.56 points on FLAIR, 4.73 points on FloodNet, 7.80 points on Potsdam, 7.60 points on Vaihingen, and 1.86 points on OpenEarthMap. This result is consistent with the motivation in Section I. When appearance statistics vary substantially across remote sensing datasets due to differences in sensors, imaging conditions, seasonal patterns, and scene composition, purely appearance-driven transfer becomes less reliable. By contrast, geometry-aware intermediate cues remain more transferable because they encode vertical organization rather than dataset-specific texture. The especially large improvements on Potsdam and Vaihingen further suggest that the proposed design is particularly effective in scenes dominated by buildings, roads, and low vegetation, where height stratification provides a strong cue for semantic separation.

The class-wise results in Table III provide a more detailed view of this advantage. On FLAIR, our method obtains the best IoU on building (77.03), road (35.23), and forest (61.40), yielding the highest overall mIoU of 46.41. Notably, the gain on road is substantial, indicating that the proposed geometry-guided representation is effective for separating elongated impervious surfaces from visually similar neighboring regions. On FloodNet, our method again ranks first overall with 45.44 mIoU and achieves the best performance on building (71.96), road (43.06), and water (56.39). The improvement on road is particularly clear, exceeding CrossEarth by 19.75 points, which indicates stronger robustness under flood-related appearance disturbance. On Potsdam, the advantage becomes even more pronounced on structure-sensitive categories, with building reaching 84.30 and road reaching 52.91, far above all competing methods. On Vaihingen, our method

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TABLE V
CLASS-WISE IOU COMPARISON UNDER THE VDD-SOURCE PROTOCOL (%)

Method	Other	Wall	Road	Vegetation	Vehicle	Roof	Water	mIoU
VDD								
LSKNet-T [51]	32.10	0.00	46.15	81.33	0.00	60.84	90.94	44.48
LSKNet-S [51]	30.81	0.00	43.92	82.41	0.00	62.56	90.89	44.37
D2LS [53]	64.49	53.97	70.44	89.41	31.59	87.76	95.24	70.41
SFANet [52]	67.23	57.91	74.52	90.03	62.22	89.90	95.64	76.78
REIN [4]	63.10	59.69	71.45	90.33	62.60	82.28	95.78	75.03
CrossEarth [26]	68.14	67.49	77.88	91.23	74.34	87.80	96.41	80.47
Ours	**73.65**	**70.64**	**80.57**	**91.65**	**75.76**	**93.77**	**96.85**	**83.27**
UDD								
LSKNet-T [51]	16.21	0.00	27.16	71.94	0.00	40.46	/	25.96
LSKNet-S [51]	14.75	0.00	27.21	75.14	0.00	41.60	/	26.45
D2LS [53]	36.19	58.61	24.02	82.45	28.67	67.03	/	49.50
SFANet [52]	34.58	58.21	26.25	82.14	48.89	70.50	/	53.43
REIN [4]	41.18	55.49	32.13	81.28	60.19	76.96	/	57.87
CrossEarth [26]	41.04	60.51	**37.85**	84.61	**69.44**	79.67	/	62.19
Ours	**47.39**	**66.56**	34.54	**85.00**	67.70	**86.76**	/	**64.66**
UAVid								
LSKNet-T [51]	19.69	0.00	27.75	63.31	0.00	23.28	1.31	19.33
LSKNet-S [51]	21.60	0.00	30.86	63.27	0.00	22.55	1.11	19.91
D2LS [53]	38.05	55.72	51.50	79.85	28.48	44.41	28.82	46.69
SFANet [52]	35.82	50.74	53.61	79.96	51.84	46.36	27.38	49.39
REIN [4]	49.56	72.78	**69.63**	79.24	67.72	73.48	60.29	67.53
CrossEarth [26]	52.40	74.65	68.91	83.89	**75.56**	**74.32**	61.80	70.22
Ours	**53.06**	**75.08**	69.25	**83.91**	69.88	74.20	**68.43**	**70.54**
SkyScapes-Dense								
LSKNet-T [51]	3.24	/	4.64	57.68	0.00	30.60	/	19.23
LSKNet-S [51]	1.61	/	5.93	62.57	0.00	33.27	/	20.67
D2LS [53]	4.63	/	25.10	74.98	8.96	39.87	/	30.71
SFANet [52]	4.19	/	23.01	76.83	17.86	50.38	/	34.45
REIN [4]	6.17	/	39.72	79.77	35.89	69.31	/	46.17
CrossEarth [26]	6.83	/	40.65	82.90	36.62	69.49	/	47.30
Ours	**7.12**	/	**44.56**	**85.77**	**41.21**	**73.61**	/	**50.45**
AeroScapes								
LSKNet-T [51]	1.24	/	14.17	49.39	0.00	20.77	/	17.11
LSKNet-S [51]	1.07	/	13.33	46.11	0.00	20.70	/	16.24
D2LS [53]	2.39	/	7.20	80.75	13.94	**31.84**	/	27.22
SFANet [52]	1.38	/	9.23	54.92	28.89	23.86	/	23.65
REIN [4]	2.43	/	28.28	79.16	48.40	7.54	/	33.16
CrossEarth [26]	1.94	/	29.14	65.36	29.72	16.43	/	28.52
Ours	**2.77**	/	**29.89**	**95.83**	**71.64**	8.69	/	**41.77**

achieves 88.59 on building and 62.26 on road, producing the highest overall mIoU of 42.67. Overall, the largest gains are concentrated on categories such as buildings, roads, and water boundaries, which is well aligned with our central claim that overhead segmentation benefits from geometry that reflects stable vertical structure. By contrast, gains on low-profile or weakly height-distinguishable categories such as agricultural are smaller, which is expected because these classes provide less distinctive geometric contrast.

The qualitative comparisons in Fig. 3 provide visual evidence for these quantitative improvements on broader remote sensing datasets. In the first and third rows, competing methods either produce noisy scattered predictions or fail to recover coherent water and vegetation regions under strong appearance variation, whereas our method yields smoother and more homogeneous masks that are closer to the ground truth. In the second row, our method better resolves the adjacency between the road structure and the water body, with cleaner linear boundaries and fewer spurious fragments. In the fourth and fifth rows, which contain complex urban layouts with roads, roofs, and impervious surfaces, our predictions preserve the global structure more clearly and reduce region bleeding across adjacent categories. In the last two rows, our method also produces cleaner large-area predictions for buildings and vegetation, suppressing isolated false positives that remain visible in prior methods. These exam-

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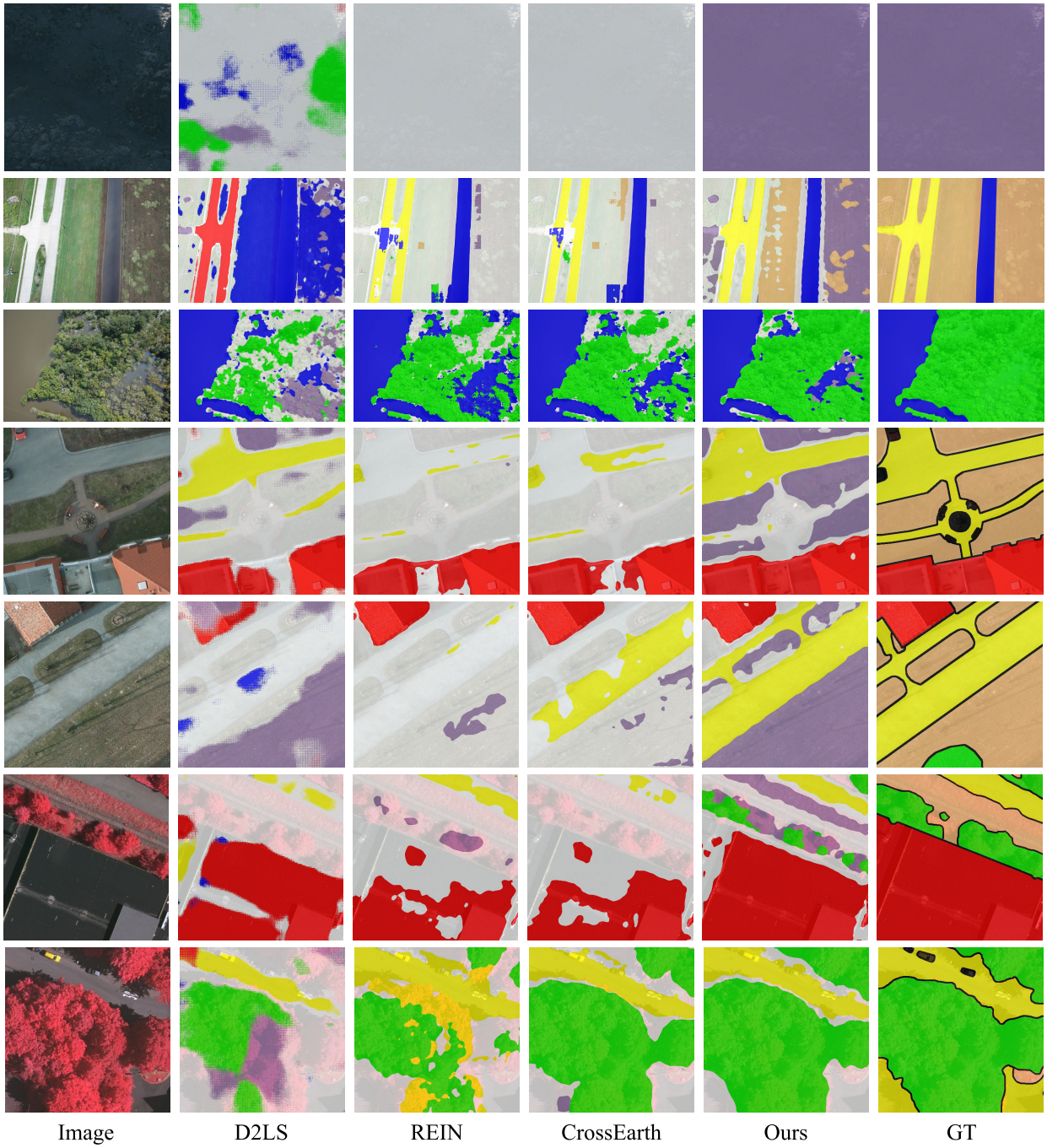


Fig. 3. Qualitative comparison under the LoveDA-source protocol. GT denotes the ground-truth mask.

ples suggest that the proposed geometry-guided representation improves both semantic consistency within homogeneous regions and boundary localization across heterogeneous regions.

Table IV reports the results under the VDD-source protocol. Our method also ranks first across this within-UAV transfer setting, achieving 83.27 mIoU on VDD, 64.66 on UDD, 70.54 on UAVid, 50.45 on SkyScapes and 41.77 on AeroScapes, with the best mean performance of 62.14. Compared with the strongest prior method, the average gain is 4.40 points, including improvements of 2.80 points on VDD, 2.47 points on UDD, 0.32 points on UAVid, 3.15 points on SkyScapes and 8.61 points on AeroScapes. On the three originally reported datasets, namely, the VDD source domain itself together with the two original unseen domains, the margins are smaller for the two protocols, because the VDD-source setting is already a strong and comparatively saturated UAV benchmark in which all methods benefit from closer domain proximity and the performance ceiling is high. The two additional unseen UAV domains sit further from the source distribution, and the margins there are wider than on any of the three originally reported datasets, although the mean gain under this protocol remains below the 4.97 points obtained under the LoveDA-source protocol. Even in this case, the proposed model still

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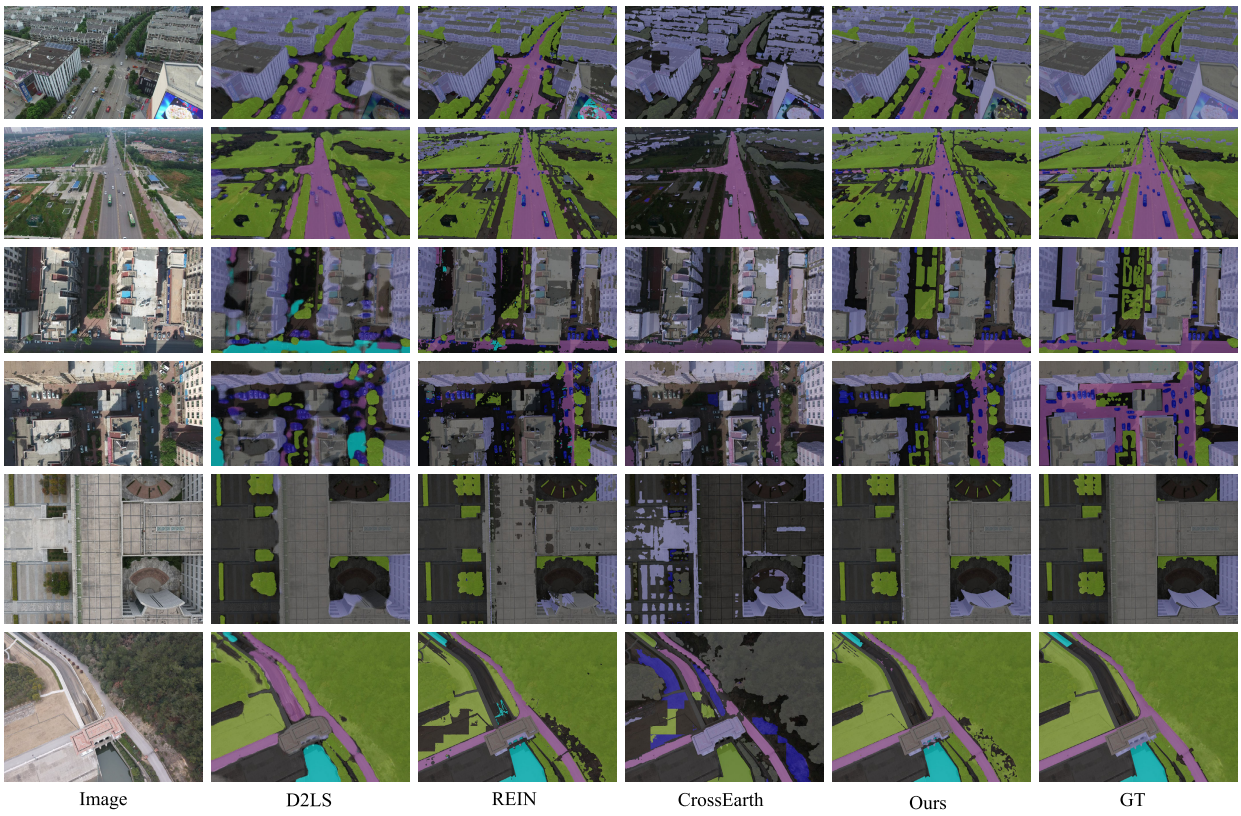


Fig. 4. Qualitative comparison under the VDD-source protocol. GT denotes the ground-truth mask.

provides consistent gains, showing that intermediate geometry serves as complementary evidence rather than merely compensating for extreme appearance gaps.

The class-wise comparison in Table V further confirms this trend. On VDD, our method achieves the best IoU in all seven categories, including other, wall, road, vegetation, vehicle, roof, and water, indicating a broad and stable improvement rather than a gain limited to a few easy classes. On UDD, our method achieves the highest overall mIoU and ranks first on other (47.39), wall (66.56), vegetation (85.00), and roof (86.76), while remaining competitive on vehicle. Although CrossEarth is slightly better on road and vehicle, our method still achieves the best overall balance across categories. On UAVid, our method again achieves the best overall mIoU and leads on other categories (53.06), wall (75.08), road (69.25), vegetation (83.91), and water (68.43). The gain on water is particularly notable, improving over CrossEarth by 6.63 points. These results indicate that the proposed method is especially effective for categories whose recognition depends not only on local appearance but also on vertical layout, spatial extent, and boundary quality.

Fig. 4 shows a similar pattern in UAV scenes. In the first two rows, our method better preserves the continuity of long road corridors and their surrounding vegetation strips, while competing methods exhibit fragmented masks or unstable responses along the road boundaries. In the third and fourth rows, which contain dense urban blocks with narrow roads and closely packed buildings, our method reduces confusion between roofs, roads, and shadowed regions, producing predictions that are more topologically coherent. In the fifth row, our method more accurately separates plaza surfaces, vegetation patches, and surrounding structures. In the last row, it also yields a cleaner decomposition of the bridge-road structure, water region, and nearby terrain. Overall, the qualitative results support the quantitative findings and further verify the design motivation in Section I: injecting intermediate geometry into a frozen VFM helps recover cross-scale context, preserve elongated structures, and sharpen boundaries under aerial domain shift.

1) Additional Unseen Domains: To test generalization beyond the domains used in the main comparison, we add three further unseen domains and evaluate all compared methods on them without any target annotation or adaptation. OpenEarthMap extends the LoveDA-source protocol with imagery from 44 countries; SkyScapes-Dense and AeroScapes extend the VDD-source protocol with a nadir airborne campaign over a German city and a low-altitude UAV campaign, respectively. Candidate benchmarks were selected on the requirement that they admit a deterministic and semantically faithful mapping into the existing unified label space, and several superficially suitable datasets were rejected because they merge buildings and roads into a single built-up class. Each new domain is evaluated at the canonical scale already used for the most similar existing target of the same protocol, fixed in advance rather than chosen from target-domain scores, and every method runs through identical pipelines and masks.

The three domains are reported as additional columns and blocks of Tables II–V. AerialToken ranks first on all three,

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TABLE VI
INFERENCE COST OF THE THREE FROZEN-VFM METHODS ON ONE RTX 4090, FOR A SINGLE 512 × 512 SLIDING WINDOW

	Params. (M)					Mean mIoU	
Method	Total	Train.	GFLOPs	ms	Mem. (GB)	LoveDA	VDD
REIN [4]	327.77	23.57	445.0	79.2	4.84	38.76	55.95
CrossEarth [26]	327.77	23.57	445.0	80.2	4.84	39.97	57.74
AerialToken (ours)	663.32	23.97	814.1	124.8	6.71	**44.94**	**62.14**

TABLE VII
COMPONENT ABLATION STUDY UNDER THE VDD-SOURCE PROTOCOL (%)

Configuration	VDD	UDD	UAVid	Mean
REIN	75.03	57.87	67.53	66.81
+ depth map (early fusion)	74.38	56.42	66.91	65.90
+ geometry-token route	83.01	64.57	70.81	72.80
+ intermediate geometry	78.65	58.74	68.92	68.77
+ dual-route calib.	81.74	59.61	70.15	70.50
+ structural decoder	80.25	59.48	69.76	69.83
AerialToken	**83.27**	**64.66**	**70.54**	**72.82**

improving on the strongest prior method by 1.86 points on OpenEarthMap, 3.15 on SkyScapes, and 8.61 on AeroScapes. The margins differ markedly between the two protocols, and the difference is informative. Under the VDD-source protocol, both new domains yield wider margins than any of the three originally reported datasets, and AeroScapes gives the widest margin of any evaluated dataset in this article. This is consistent with the analysis above: these domains introduce larger changes in sensor, altitude, and acquisition geometry, so appearance statistics transfer less well while the height-related structure that the geometry route encodes remains informative. Under the LoveDA-source protocol, the picture is the opposite. OpenEarthMap is the least difficult target of that protocol for the strongest methods, and REIN, CrossEarth and AerialToken each reach their highest LoveDA-source score there, so less is left for a geometry prior to recover, and the margin is the narrowest of the five targets. We report this because it delimits where the mechanism helps: the geometry route contributes most when the appearance gap is wide, and least when a target is already well covered by appearance cues alone. The ordering among prior methods is also not fully preserved on the new domains. REIN overtakes CrossEarth on AeroScapes although CrossEarth leads it on every other target of both protocols, and SFANet overtakes D2LS on OpenEarthMap although D2LS leads on all four original LoveDA-source targets, which indicates that these domains probe a somewhat different failure mode rather than simply repeating the original ones. The per-class pattern points the same way, with the largest single-class margins on road for OpenEarthMap (65.92 against 49.71) and on vehicle for AeroScapes (71.64 against 48.40), so gains concentrate on categories whose separation depends on spatial structure rather than on local color. The new targets are included in the means of Tables II and IV, whereas the ablation tables remain on the original target sets and are compared only within that scope.

2) Computational Cost: The geometry branch is a second frozen ViT-L, so it is not free, and Table VI reports what it costs. All three methods are measured on the same unit of work, one 512 × 512 sliding window at batch size 1 in single precision, which is directly comparable because all three use the identical window size and stride at inference and therefore process the same number of windows per image. Fused attention kernels are disabled for every method, so the comparison reflects arithmetic rather than which kernels happen to be installed. REIN and CrossEarth coincide in this table because their inference-time segmentors are the same architecture, a rank-factorized token adapter on a frozen DINOv2 ViT-L with a Mask2Former head; CrossEarth differs in how that network is trained, not in what runs at test time.

Relative to that shared baseline, AerialToken uses 2.02 times the parameters, 1.83 times the FLOPs, 1.58 times the latency, and 1.39 times the peak activation memory, at 6.71 GB against 4.84 GB. The parameter growth is entirely frozen weight. The trained parameter count rises by only 0.41 M, from 23.57 to 23.97 M, and that increment is the structural decoder: the geometry route itself adds six trained scalars to the token adapter, namely, the two affinity temperatures, the mixing logit, the routing gate, and the two route scales. Training cost and optimizer state are therefore essentially unchanged, and the added encoder contributes no backward pass. Against this inference cost, the mean gain over the stronger of the two baselines is 4.97 mIoU under the LoveDA-source protocol and 4.40 under the VDD-source protocol. Whether that trade is worthwhile is a deployment question rather than a general one, and we note only that the added component is a standard released ViT-L held frozen, so the usual inference-time reductions apply to it independently of the rest of the design.

D. Ablation Study

Table VII reports the ablation results under the VDD-source protocol. Starting from the REIN baseline, fusing the final depth map with the image at the input stage reduces the mean mIoU from 66.81 to 65.90. This result supports our main observation that final dense depth is not a sufficiently stable standalone transferable cue for aerial domain generalization.

1) How the Final Depth Map Is Used: The "+ depth map (early fusion)" row combines the depth map with the RGB input before the frozen visual encoder, so it differs from the proposed design in both the geometry representation and the point at which that representation enters the network. Because the comparison between final dense depth and intermediate geometry is central to this work, we also evaluate a token-level

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TABLE VIII

MECHANISM SENSITIVITY UNDER THE LOVEDA-SOURCE PROTOCOL (%). EACH ROW IS RETRAINED IN A CONTROLLED SENSITIVITY SUITE; AND VALUES ARE USED AS WITHIN-SUITE TRENDS RATHER THAN AS THE FINAL MAIN-COMPARISON CHECKPOINT

Variant	FLAIR	FloodNet	Potsdam	Vaihingen	Mean
(A) Encoder blocks receiving geometry calibration					
Early blocks only (0–7)	44.32	41.49	27.59	34.66	37.02
Middle blocks only (8–15)	45.65	43.15	28.73	35.39	38.23
Late blocks only (16–23)	44.79	44.20	29.85	40.30	39.79
All blocks (ours)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
Final depth map in place of intermediate tokens	43.14	43.77	29.65	39.68	39.06
Final depth map, capacity-matched	44.89	43.33	29.39	35.14	38.19
(B) Token bank size M					
$M = 50$	43.80	44.49	30.22	36.86	38.84
$M = 100$ **(ours)**	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
$M = 200$	29.02	24.34	14.28	12.64	20.07
$M = 200$, queries held at 100	43.80	44.08	29.38	38.38	38.91
(C) Mixing coefficient α initialization					
Visual-biased ($\alpha_0 \approx 0.05$)	44.53	43.93	28.66	37.09	38.55
Default (ours) ($\alpha_0 \approx 0.27$)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
Geometry-biased ($\alpha_0 \approx 0.73$)	35.79	28.90	20.71	22.10	26.88
(D) Relative clipping					
Disabled	43.43	43.33	28.84	33.88	37.37
Enabled (ours)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
(E) Cross-layer query aggregation					
Last-layer only	43.76	43.81	28.78	34.83	37.80
Cross-layer (ours)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
(F) Top-ρ geometry masking					
Disabled	44.21	44.42	28.84	37.63	38.78
Enabled (ours)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
(G) Structure/boundary decoder branches					
w/o structure/boundary	45.74	44.78	28.78	36.74	39.01
w/o structure branch only	44.17	45.20	29.10	36.95	38.86
w/o boundary branch only	45.09	43.87	29.28	39.01	39.31
Full design (ours)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**
(H) Geometry signal versus encoder capacity					
Depth-encoder backbone, geometry route off	44.74	43.67	28.77	37.43	38.65
Full design (ours)	**47.36**	**45.55**	**30.26**	**38.11**	**40.32**

counterpart that isolates the representation alone, reported in Table VIII, and we specify it exactly. The input image is bilinearly resized to 512×512 and passed through the same frozen Depth Anything V2 model, whose *final* decoder output is taken as a single-channel relative depth map. The map is bilinearly resized to the visual token grid $(H/16, W/16)$ and standardized per image by subtracting its spatial mean and dividing by its spatial standard deviation, clamped from below at 10^{-6}; this per-image normalization is what removes the arbitrary global scale of relative monocular depth, and no metric calibration is applied. The normalized map is projected from one channel to $C = 1024$ by a single trainable linear layer, a zero-valued class token is prepended, and the resulting sequence replaces G^k at *every* encoder block. The fusion point, the token bank, the affinity computation, the top-ρ screening, the relative clipping, and the residual injection are all identical to the full model; only the source of the geometry token changes. Two settings of this counterpart are reported: one that disables the bank-derived queries and the decoder refinements, and a capacity-matched one that retains them, so that the geometry source is the only quantity that changes with respect to the corresponding intermediate-geometry row. The depth-map projector is the only additional trainable parameter introduced, so the variant is not disadvantaged by reduced capacity.

The geometry-token route row is the full model with the cross-scale and boundary refinements disabled. It reaches 72.80 against a full-model anchor of 72.72 retrained under the identical recipe, so on this protocol those two refinements do not measurably contribute. Under the LoveDA-source protocol, the same two branches are worth 1.31 points within the sensitivity suite of Table VIII, group (G), so they matter more as the domain shift widens.

In contrast, routing intermediate geometry through the alignment of Section III-B improves the mean mIoU to 68.77, showing that intermediate geometric representations are more effective than the raw depth map for cross-domain transfer. Based on this aligned geometry cue, layerwise dual route token calibration further boosts the mean mIoU to 70.50, indicating that explicitly balancing appearance-driven and geometry-driven token interactions is important. The query-conditioned structural decoder also improves the performance to 69.83, suggesting that geometry-aware query refinement and structural mask decoding bring additional benefits at the decoding stage.

When all components are combined, AerialToken achieves the best performance of 72.82 mean mIoU, reaching 83.27 on VDD, 64.66 on UDD, and 70.54 on UAVid. Compared with the REIN baseline, the full model improves the mean by 6.01 points. These results consistently verify that the main gain does not come from directly using the final depth map, but from exploiting aligned intermediate geometry and carrying it through calibrated fusion and structure-aware decoding.

2) LoveDA-Source Component Ablation: Table I extends the component analysis to the LoveDA-source protocol and

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TABLE IX

INFERENCE-TIME DIAGNOSTIC ABLATIONS UNDER THE LOVEDA-SOURCE PROTOCOL (%). ROWS SWITCH ROUTING OR FEATURE SELECTION ON THE SAME FIXED MAIN-COMPARISON CHECKPOINT. BOLD DENOTES THE PRESPECIFIED DEFAULT USED IN THE MAIN COMPARISONS, NOT POST-HOC TARGET-DOMAIN SELECTION

Variant	FLAIR	FloodNet	Potsdam	Vaihingen	Mean
(A) Affinity routing					
Geometry-only	46.85	45.82	36.53	45.13	43.58
Dual-route (ours)	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**
Visual-only	46.58	45.55	35.53	42.99	42.66
(B) Geometry feature depth					
Early layers	45.92	46.09	33.66	44.86	42.63
Middle layers	45.84	45.09	32.76	45.00	42.17
Late layers	48.65	46.59	35.83	46.55	44.41
All blocks (ours)	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**
(C) Relative clipping radius r					
Disabled	4.38	3.78	4.58	5.60	4.59
$r = 0.05$	48.48	46.43	35.47	46.88	44.32
$r = 0.10$ **(ours)**	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**
$r = 0.20$	34.30	37.29	31.56	33.18	34.08
$r = 0.50$	3.77	6.68	4.82	0.16	3.86
(D) Query aggregation					
Mean pooling	22.21	22.16	9.06	14.37	16.95
Last-layer only	44.35	44.27	32.55	39.35	40.13
Cross-layer (ours)	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**
(E) Top-ρ geometry masking					
$\rho = 0.10$	46.40	45.43	35.56	42.61	42.50
$\rho = 0.25$ **(ours)**	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**
$\rho = 0.50$	46.45	45.46	35.54	42.76	42.55
$\rho = 0.75$	46.51	45.49	35.48	42.87	42.59
Disabled ($\rho = 1$)	46.57	45.55	35.45	43.06	42.66
(F) Decoder branches					
w/o structure branch	46.68	45.68	35.17	43.13	42.67
w/o boundary branch	46.41	45.44	35.57	42.67	42.52
Full design (ours)	**46.41**	**45.44**	**35.57**	**42.67**	**42.52**

TABLE X

CROSS-DOMAIN DISCREPANCY OF THE TWO GEOMETRY REPRESENTATIONS UNDER THE LOVEDA-SOURCE PROTOCOL. VALUES ARE MMD^2 NORMALIZED BY A WITHIN-DOMAIN NULL (1.0 = TWO HALVES OF THE SAME DATASET); AND LOWER IS MORE STABLE. BOLD MARKS THE MORE STABLE REPRESENTATION IN EACH PAIR

	Raw		Model-normalized	
Domain pair	Geometry	Depth map	Geometry	Depth map
LoveDA ↔ FLAIR	38.03	**18.99**	37.89	**23.74**
LoveDA ↔ FloodNet	**42.57**	54.88	45.49	**12.77**
LoveDA ↔ Potsdam	**40.05**	143.54	40.25	**23.84**
LoveDA ↔ Vaihingen	**57.76**	275.69	**57.73**	95.59
FLAIR ↔ FloodNet	37.90	**13.43**	44.71	**5.95**
FLAIR ↔ Potsdam	**30.84**	62.44	29.68	**3.55**
FLAIR ↔ Vaihingen	**67.34**	156.71	66.15	**27.41**
FloodNet ↔ Potsdam	38.60	**32.14**	43.45	**12.92**
FloodNet ↔ Vaihingen	**58.59**	99.15	57.26	**55.20**
Potsdam ↔ Vaihingen	57.20	**21.82**	58.16	**28.24**
Mean	**46.89**	87.88	48.08	**28.92**
Pairs favouring geometry	**6 / 10**		1 / 10	
Mean CORAL	0.01327	**0.00937**	0.01352	**0.00574**

aligns the REIN and AerialToken rows with the main class-wise results in Table III. The full AerialToken model achieves 42.52 mean mIoU (FLAIR 46.41, FloodNet 45.44, Potsdam 35.57, and Vaihingen 42.67). The REIN baseline reaches 35.45 mean mIoU, and fusing the final depth map with the image at the input stage reduces the mean to 35.17, confirming that the final dense depth prediction alone is not consistently reliable as a standalone transferable cue. That row changes the geometry representation and the point of fusion at the same time, so the representation itself is isolated by two further comparisons in which the geometry source is the only quantity that varies. Injecting the same final depth map as tokens through the calibration path, rather than at the input, reaches 39.06 in Table VIII against 39.77 for the intermediate tokens in the present table under otherwise identical settings, and the capacity-matched pair, which additionally retains the bank-derived queries and the decoder refinements, reaches 38.19 against 40.32. The deficit is therefore attributable to the representation rather than to reduced capacity or to a weaker fusion point, and it widens as the rest of the design is restored, which is what makes the parameter-free pairing of Section III-B the operative design choice rather than an implementation convenience. The geometry-token route row of the present table is the full model with the cross-scale and boundary refinements disabled. Because the alignment introduces no parameters, the "+ intermediate geometry" row is a substitution of the geometry representation rather than the insertion of a module. The ablation rows for intermediate geometry, dual-route calibration, and the structural decoder report 39.77, 39.53, and 38.60 mean mIoU, respectively. Their gaps to the full model confirm that the gain in AerialToken comes from the combined use of aligned intermediate geometry, dual-route token calibration, and structure-aware decoding rather than from adding a raw depth map.

All ablation tables in this section are computed over the four original LoveDA-source targets and therefore use a narrower scope than Table II, whose mean now also includes OpenEarthMap. The reference value for the ablations is accordingly 42.52 rather than the 44.94 reported in the main comparison, and the same applies on the VDD-source side, where the ablation reference is 72.82 over the three originally reported

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TABLE XI

RAW-TO-UNIFIED LABEL REMAPPING USED IN THE LOVEDA-SOURCE PROTOCOL. LABEL 0 DENOTES BACKGROUND OR IGNORED CATEGORIES AND IS EXCLUDED FROM CLASS-WISE IOU REPORTING

Dataset	Raw id(s)	Raw class(es)	Remapping rule	Unified id	Unified class
LoveDA					
LoveDA	0	no-data	ignored	0	Ignored
LoveDA	1	background	mapped to evaluation background	0	Background / ignored
LoveDA	2	building	direct match	1	Building
LoveDA	3	road	direct match	2	Road
LoveDA	4	water	direct match	3	Water
LoveDA	5	barren	direct match	4	Barren
LoveDA	6	forest	direct match	5	Forest
LoveDA	7	agriculture	direct match	6	Agricultural
FLAIR					
FLAIR	1	building	direct match	1	Building
FLAIR	3	impervious surface	merged into the road class	2	Road
FLAIR	5	water	direct match	3	Water
FLAIR	4	bare soil	direct match	4	Barren
FLAIR	6,7,8,16,17	coniferous, deciduous, brushwood, ligneous, mixed	merged into woody vegetation	5	Forest
FLAIR	9,11,12	vineyard, agricultural land, plowed land	merged into agricultural surfaces	6	Agricultural
FLAIR	2,10,13,14,15, 18,19	pervious surface, herbaceous vegetation, swimming pool, snow, clear-cut, greenhouse, other	ignored	0	Ignored
FloodNet					
FloodNet	1	background	mapped to evaluation background	0	Background / ignored
FloodNet	2,3	building flooded, building non-flooded	merged into building	1	Building
FloodNet	4,5	road flooded, road non-flooded	merged into road	2	Road
FloodNet	6	water	direct match	3	Water
FloodNet	7	tree	merged into forest	5	Forest
FloodNet	10	grass	merged into agricultural land	6	Agricultural
FloodNet	8,9	vehicle, pool	ignored	0	Ignored
Potsdam and Vaihingen					
Potsdam/Vaihingen	2	building	direct match	1	Building
Potsdam/Vaihingen	1	impervious surfaces	merged into road	2	Road
Potsdam/Vaihingen	4	tree	merged into forest	5	Forest
Potsdam/Vaihingen	3	low vegetation	merged into agricultural land	6	Agricultural
Potsdam/Vaihingen	5,6	car, clutter/background	ignored	0	Ignored
OpenEarthMap					
OpenEarthMap	8	building	direct match	1	Building
OpenEarthMap	4	road	direct match	2	Road
OpenEarthMap	6	water	direct match	3	Water
OpenEarthMap	1	bareland	direct match	4	Barren
OpenEarthMap	5	tree	merged into woody vegetation	5	Forest
OpenEarthMap	7	agriculture land	direct match	6	Agricultural
OpenEarthMap	2,3	rangeland, developed space	LoveDA assigns the equivalent content to its own background class, so no foreground counterpart exists	0	Ignored
OpenEarthMap	0	unknown	ignored	0	Ignored

datasets against 62.14 over five. The ablation variants were not re-evaluated on the additional domains, so we keep them on the original evaluation sets and compare within each scope only.

The LoveDA-source ablation tables use complementary numerical scopes. Tables I and IX are tied to the final main-comparison checkpoint in Table III, whose reference mean is 42.52. Table VIII, by contrast, is a controlled trained-sensitivity suite: each design choice is retrained under the same ablation recipe, and the default row in that suite reaches 40.32 mean mIoU. We therefore use Table VIII to compare relative changes within the sensitivity suite, not to reselect a final model or to replace the main-comparison checkpoint.

3) Quantifying Cross-Domain Stability of the Two Geometry Representations: The claim that intermediate geometry is a more stable cross-domain cue than the final dense depth map should not rest on segmentation accuracy alone, so we measure it directly. For each dataset, we draw 200 images, pass them through the same frozen Depth Anything V2 model, and form two per-image descriptors: the spatially pooled intermediate tokens, and the dense depth map resampled to the token grid. Both are reduced to the same dimensionality by a principal component analysis (PCA) fit on the pooled set across all domains, so the comparison is not confounded by native dimensionality, and each between-domain squared maximum mean discrepancy (MMD^2) is normalized by a within-domain null obtained from random half-splits of the same dataset. No trained segmentation checkpoint enters this measurement.

Read together with the segmentation results, Table X supports a sharper claim than an unconditional one: the advantage of intermediate geometry appears exactly when the global relative-depth scale actually drifts across the domains in question. Under the LoveDA-source protocol, which spans satellite and airborne sensors with markedly different ground sampling distances, the raw dense depth map is highly unstable across domains (mean normalized MMD^2 87.88) while the

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TABLE XII

LABEL REMAPPING USED IN THE VDD-SOURCE PROTOCOL. THE UNIFIED LABEL SPACE FOLLOWS THE SEVEN-CLASS VDD TAXONOMY. UAVID ENTRIES FOLLOW THE VDD AUTHORS' IDD-COMPATIBLE ANNOTATIONS; AND NO NEW PIXEL ANNOTATION IS INTRODUCED IN THIS WORK

Dataset	Raw id(s)	Raw class(es)	Remapping rule	Unified id	Unified class
VDD					
VDD	0	other	direct match	0	Other
VDD	1	wall	direct match	1	Wall
VDD	2	road	direct match	2	Road
VDD	3	vegetation	direct match	3	Vegetation
VDD	4	vehicle	direct match	4	Vehicle
VDD	5	roof	direct match	5	Roof
VDD	6	water	direct match	6	Water
UDD6 (original masks used here)					
UDD6	0	other	direct match	0	Other
UDD6	1	facade	remapped to wall	1	Wall
UDD6	2	road	direct match	2	Road
UDD6	3	vegetation	direct match	3	Vegetation
UDD6	4	vehicle	direct match	4	Vehicle
UDD6	5	roof	direct match	5	Roof
UDD6	N/A	no water category in the original masks used here	absent from this target label set	6	Water (not evaluated)
UAVid (VDD/IDD-compatible annotations)					
UAVid	0	other/clutter	remapped to other	0	Other
UAVid	1	wall	direct match	1	Wall
UAVid	2	road	direct match	2	Road
UAVid	3	vegetation	direct match	3	Vegetation
UAVid	4	vehicle	direct match	4	Vehicle
UAVid	5	roof	direct match	5	Roof
UAVid	6	water	direct match	6	Water
SkyScapes-Dense					
SkyScapes	2,3,4,5,6,7,8,9,10	paved/non-paved road, paved/non-paved parking, bikeways, sidewalks, entrance exit, danger area, lane-markings	merged into road as the closest hard-surface category	2	Road
SkyScapes	1,20	low vegetation, tree	merged into vegetation	3	Vegetation
SkyScapes	12–17	car, trailer, van, truck, long truck, bus	merged into vehicle	4	Vehicle
SkyScapes	11	building	nadir view: mapped to roof	5	Roof
SkyScapes	18	clutter	merged into other	0	Other
SkyScapes	19	impervious surface	spans low wall, rocky terrain and river, so no deterministic mapping exists	–	Ignored
SkyScapes	N/A	no wall or water category	absent from this target label set	1, 6	Wall, Water (not evaluated)
AeroScapes					
AeroScapes	10	road	direct match	2	Road
AeroScapes	9	vegetation	direct match	3	Vegetation
AeroScapes	2,3,5	bike, car, boat	merged into vehicle	4	Vehicle
AeroScapes	8	construction	mapped to roof	5	Roof
AeroScapes	1,4,6,7	person, drone, animal, obstacle	merged into other	0	Other
AeroScapes	0,11	unlabeled, sky	sky has no counterpart in an overhead label space	–	Ignored
AeroScapes	N/A	no wall or water category	absent from this target label set	1, 6	Wall, Water (not evaluated)

intermediate tokens are not (46.89), and the intermediate representation is the more stable of the two on 6 of 10 domain pairs. The instability is not uniform: it reaches 275.69 on LoveDA–Vaihingen, 156.71 on FLAIR–Vaihingen and 143.54 on LoveDA–Potsdam, and stays moderate elsewhere. Potsdam and Vaihingen are precisely the two targets on which AerialToken obtains its largest margins (+7.80 and +7.60 over the strongest prior method), so the representation-level and the task-level measurements localize the same effect to the same datasets.

Under the VDD-source protocol, the picture differs, and we report it rather than omit it. Those domains are UAV acquisitions at broadly comparable altitudes, and the raw dense depth map is already far more stable across them than across the LoveDA-source domains (13.33 versus 87.88), leaving the intermediate tokens no clear advantage: the two representations trade places across pairs, with a mean of 15.96 for geometry against 13.33 for the dense map. The corresponding segmentation margin on the original VDD-source targets is also the smaller of the two protocols, at +1.86 against +5.74 before the additional domains are included. The two protocols therefore agree on a single reading: the magnitude of the geometry advantage tracks the magnitude of depth-scale drift in the domain shift, which is what the design predicts, since an intermediate representation can only improve on a dense prior that is itself unreliable. This delimits where the method should be expected to help, namely, under heterogeneous sensors, altitudes, and ground sampling distances, and where the appearance path already suffices.

Two caveats are stated explicitly. First, in the model-normalized regime the ordering reverses on the LoveDA-source protocol (28.92 versus 48.08), because the per-image

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standardization that the "+ depth map" variant applies is exactly an operation that removes the drifting global scale. That stability is bought by removing global relative-depth scale and offset cues, which is why the standardized depth-map variant still underperforms the intermediate-token route in mIoU (38.19 versus 40.32 in the capacity-matched row of Table VIII); the intermediate tokens require no such normalization to be stable. Second, correlation alignment (CORAL), which compares only second-order statistics, favors the dense map in every setting we measured. The two discrepancy measures therefore do not agree, and we do not claim that intermediate geometry dominates under all of them.

4) LoveDA-Source Mechanism Analysis: Under this controlled trained-sensitivity scope, Table VIII examines sensitivity to individual design choices within the full model; each row is a separately trained variant. (A) Calibrating every encoder block is the default setting and anchors groups (B)–(G), reaching 40.32 mean mIoU (FLAIR 47.36, FloodNet 45.55, Potsdam 30.26, and Vaihingen 38.11). Restricting calibration to a contiguous third of the encoder is consistently worse in the mean: early blocks reach 37.02, middle blocks 38.23, and late blocks 39.79. The ordering early < middle < late indicates that deeper geometry representations carry more transferable structure, which is consistent with the inference-time analysis in Table IX. Two per-dataset observations qualify this trend. First, late-block-only calibration is the strongest variant on Vaihingen (40.30 versus 38.11 for the default), the dataset in which height stratification is most discriminative, while the default remains better on the other three datasets and in the mean; using all blocks therefore trades a small amount of Vaihingen accuracy for a more uniform gain elsewhere. Second, replacing the intermediate tokens with the final depth map reaches 39.06 mean, below both the default (40.32) and late-block-only calibration (39.79); its relative advantage on Vaihingen (39.68) does not carry over to FLAIR (43.14) or FloodNet (43.77). Giving the dense map the same treatment as the intermediate tokens does not close this gap but widens it: the capacity-matched row retains dual-route calibration, geometry-conditioned queries, and the structural decoder and changes only the source of the geometry token, and it reaches 38.19 against 40.32 for the default. The deficit therefore lies in the representation itself rather than in the reduced decoder used by the simpler depth-map row. Taken together, this group indicates that the benefit arises from intermediate geometry distributed across the encoder rather than from any single depth range or from the final dense prediction. (B) For token bank size M, the default $M = 100$ achieves the best result within the sensitivity suite. A smaller bank ($M = 50$, 38.84) reduces performance, with the largest drop on FLAIR (43.80 versus 47.36). An oversized bank ($M = 200$, 20.07) causes severe degradation across all datasets (FLAIR 29.02, FloodNet 24.34, Potsdam 14.28, and Vaihingen 12.64). This collapse is not a capacity limit. In our implementation, the token bank also determines the number of decoder mask queries, so enlarging it changes two quantities at once. Holding the query count at 100 while keeping the bank at 200 recovers the mean to 38.91, about 93% of the drop, which identifies decoder query over-parameterization as the dominant cause: with 200 queries over a label space of at most seven classes, the set-prediction assignment degenerates. The residual 1.41-point gap to the default is the cost attributable to the larger bank itself. (C) For the mixing coefficient α, geometry-biased initialization leads to highly unstable training (mean 26.88, with Potsdam collapsing to 20.71 and Vaihingen to 22.10). Visual-biased initialization is also inferior to the default by 1.77 points. (D) Relative clipping is critical for training stability: disabling it reduces the mean from 40.32 to 37.37, with a particularly large drop on Vaihingen (−4.23 pts); at inference time, disabling clipping on the main checkpoint causes a catastrophic collapse to 4.59 mean (see Table IX), confirming that the clipping constraint is structurally load-bearing across both scopes. (E) Cross-layer query aggregation contributes 2.52 points over last-layer-only aggregation, including a gain of 3.28 points on Vaihingen. (F) Top-ρ geometry masking provides a consistent 1.55-point benefit across all datasets by suppressing unreliable geometry positions during calibration. (G) The structure and boundary decoder branches together contribute 1.31 points in mean mIoU. Removing them individually shows that the two are not independent. Relative to the configuration with neither branch (39.01), the structure branch alone gives 39.31 (+0.30), and the boundary branch alone gives 38.86 (−0.15), while the pair gives 40.32 (+1.31). The joint effect is therefore substantially larger than the sum of the individual ones, which suggests a positive interaction of about 1.16 points. As these are single runs without repeated seeds, we read this as evidence of an interaction rather than as an established quantity. It also explains the observation that the boundary branch contributes little: on its own it does, and its benefit appears only in combination with the structure branch. Because every row in this suite comes from a single run, we treat large degradations only as qualitative evidence of sensitivity or failure modes under the evaluated setting, rather than as precise estimates of effect size. We do not interpret small numerical differences as statistically established improvements. (H) Replacing the frozen DINOv2 backbone with the frozen depth encoder and switching the geometry route off leaves the model with a single ViT-L of identical size and reaches 38.65, that is 1.67 points below the full design. The gain therefore does not follow from the presence of a second large encoder; it requires the geometry signal to be routed into the visual representation.

5) LoveDA-Source Inference-Time Diagnostics: Table IX probes the fixed final main-comparison AerialToken checkpoint without retraining. The reference mean is 42.52, with FLAIR 46.41, FloodNet 45.44, Potsdam 35.57, and Vaihingen 42.67. The reference row is the prespecified inference setting used in all main comparisons, while the switched rows are diagnostic probes and are not used for post-hoc model selection from target-domain scores. This explains why some diagnostic rows, such as geometry-only routing and late-layer switching, can be numerically higher than the default row while still not replacing the reported main comparison. The main model remains the prespecified configuration, and Table IX is used only to diagnose which pathway carries transferable signals. The most critical finding is that *relative clipping is structurally load-bearing*: disabling it at inference collapses all four datasets to near-zero performance (mean

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4.59, FLAIR 4.38, FloodNet 3.78, Potsdam 4.58, and Vaihingen 5.60), confirming that the clipping bound is essential to stable geometry-token modulation, not a marginal regularizer. *Cross-layer query aggregation* is likewise critical: switching to last-layer-only tokens reduces the mean to 40.13 (−2.39 pts), and mean-only pooling causes near-total failure (16.95 mean). *Geometry routing*: the geometry-only affinity route (43.58 mean) exceeds the default dual-route inference setting by 1.06 pts, with the largest changes on Potsdam (+0.96) and Vaihingen (+2.46). We interpret this as evidence that the trained geometry branch contains a strong transferable signal, rather than as a new selected final configuration. *Geometry feature depth*: switching to late intermediate layers at inference (44.41 mean, +1.88 pts) provides a diagnostic trend consistent with the trained-ablation finding in Table VIII group (A), indicating that late intermediate geometry carries richer structural cues. Early and middle layer switching provide only modest changes (+0.11 and −0.35, respectively). *Decoder branches* have negligible inference-time effect: removing the structure branch yields 42.67 (+0.15), and removing the boundary branch yields 42.52 (±0.00 relative to full model), which indicates that their main contribution arises from how they shape optimization during training rather than from their immediate test-time activation. *Top-ρ masking* similarly has minimal inference-time effect (42.66 without versus 42.52 default). The screening operator is recomputed in every forward pass and is therefore active at test time; what this probe shows is that disabling it on an already-trained checkpoint costs little, not that it is inactive during inference.

6) Sensitivity to r and ρ, and Why the Defaults Are Not Reselected: Groups (C) and (E) of Table IX sweep the two hyperparameters whose sensitivity was not previously reported. Both are recomputed inside every forward pass, so they can be varied on the fixed main-comparison checkpoint. The clipping radius gives 44.32 at $r = 0.05$, 42.52 at the default $r = 0.10$, 34.08 at $r = 0.20$ and 3.86 at $r = 0.50$, with 4.59 when clipping is removed entirely. Performance therefore declines progressively as the radius grows beyond 0.05, and both $r = 0.50$ and the unclipped setting approach collapse. Two observations make a purely discontinuous numerical artifact less likely: the degradation is graded over an order of magnitude in r rather than appearing only at a single switch, and a variant retrained without clipping converges normally to 37.37 instead of failing. This is the behavior expected of an unbounded residual accumulating across the 24 encoder blocks. The screening ratio behaves in the opposite way: across $\rho \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$ the mean varies only between 42.50 and 42.66, a total spread of 0.16 points. This indicates that a fixed ρ is adequate and that an adaptive schedule is not required at inference. Although the operator remains active at inference, the retraining ablation indicates that most of its contribution is realized through the learned training trajectory: retraining without it costs 1.55 points [Table VIII, group (F)].

Three configurations across Tables VIII and IX score above the default on the target domains, and we treat them together because they behave identically once examined. Retraining the geometry-only affinity route under the main recipe gives 42.79 against 42.52 for the default; retraining with calibration restricted to the late encoder blocks gives 43.83; and tightening the clipping radius to $r = 0.05$ gives 44.32. On the LoveDA source split, the ordering is exactly reversed, at 50.61 for the default against 49.88, 49.09, and 48.74, respectively. Target-domain and source-domain accuracy are strictly anti-correlated across these four settings: every step that buys accuracy on the unseen targets costs accuracy on the source. Because the source split is the only data that a source-only protocol permits for model selection, every admissible selection rule returns the default configuration, and adopting any of the alternatives would amount to selecting on the test domains. We therefore report the measurements and keep the default. More generally, none of the hyperparameters in Section IV were tuned: they were fixed once before the experiments and left unchanged, so the reported configuration is not the product of a search.

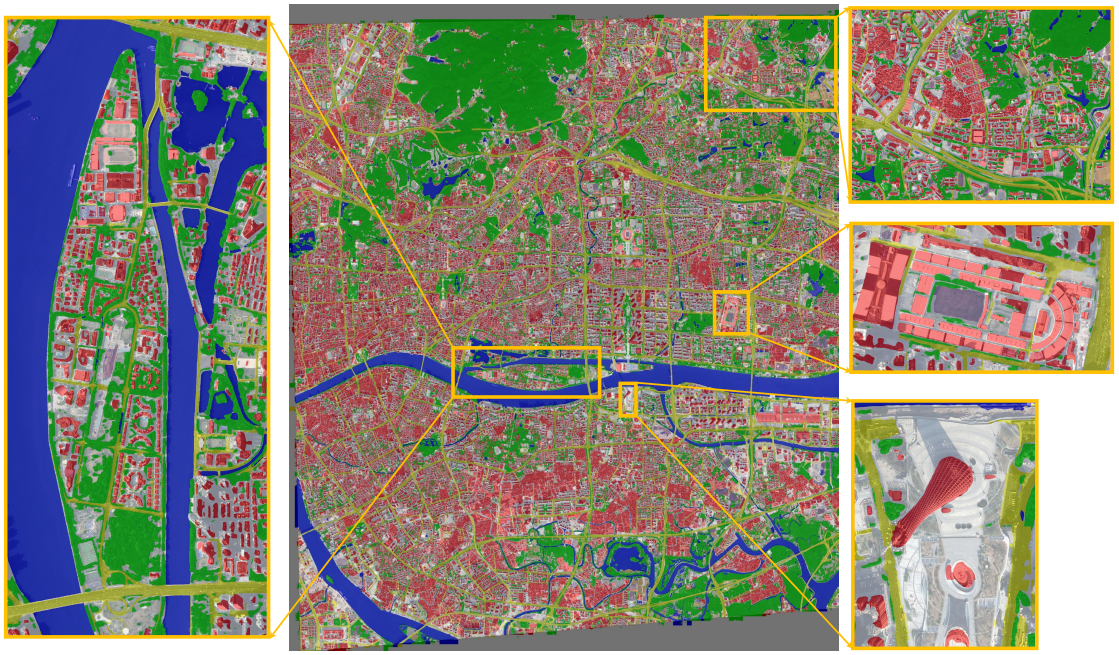
E. Qualitative Analysis

We further visualize zero-shot predictions on representative Guangzhou scenes as qualitative evidence of how the proposed geometry-guided design behaves under real cross-domain variation. Consistent with the discussion in Section I, the benefit of AerialToken is not only reflected in higher quantitative scores, but also in stronger structural coherence, finer local masks, and more stable recognition under large viewpoint and dataset shifts.

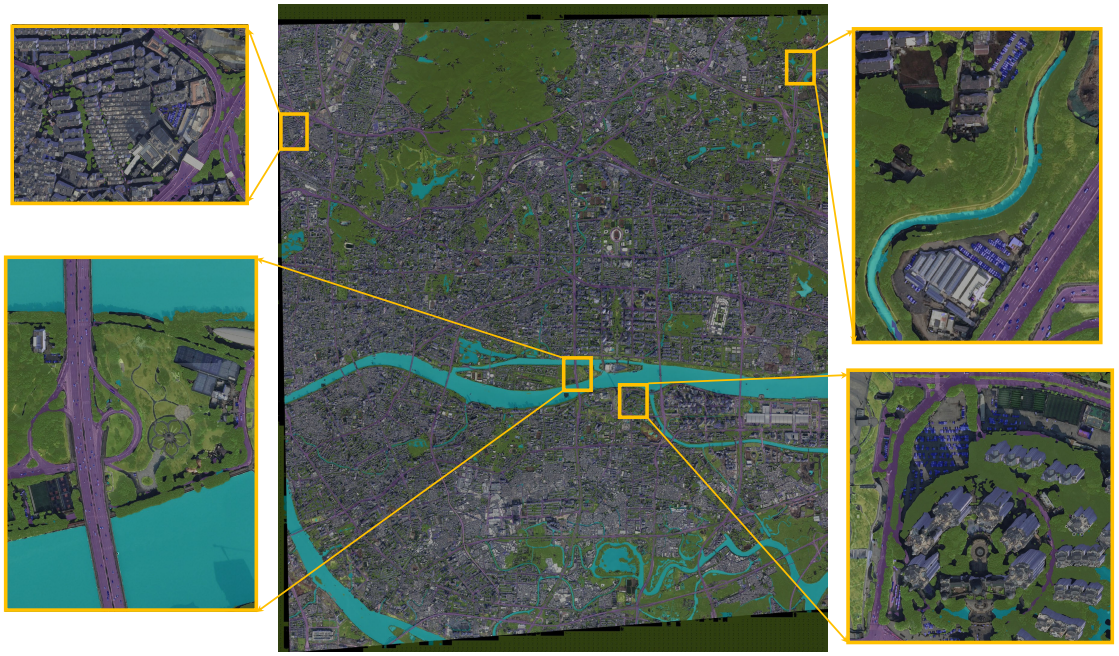
Fig. 5 provides a city-scale comparison in both satellite and UAV settings. In the satellite panel, AerialToken preserves the overall spatial organization of the urban core, including long river corridors, dense built-up regions, and large vegetation areas, while still remaining locally discriminative in the zoomed-in-view regions. A particularly striking case is the Guangzhou Tower, which is correctly recognized as a *building* even though such a slender and highly nonrectilinear building shape does not exist in the training data. This result suggests that the model is not merely matching familiar roof textures, but can transfer a more geometry-aware notion of built structures. In the UAV panel, the predictions are also notably fine-grained. Small vehicles remain visible as compact semantic responses, building footprints stay clear in dense urban blocks, and road masks remain continuous across wide roads, ramps, and curved corridors. These examples indicate that the proposed representation supports both large-scale layout understanding and detailed urban parsing.

Finally, the attention visualizations in Fig. 6 provide feature-level evidence for the same trend. In both satellite and UAV cases, the attention maps of AerialToken are less diffuse and more consistently aligned with semantically meaningful structures such as shorelines, road corridors, and building masses, whereas the baseline attention is more easily distracted by local appearance variation. The corresponding predictions are cleaner and more spatially coherent, and the PCA visualizations also suggest a more organized feature distribution. Overall, these qualitative results support the central argument of Section I: in overhead scenes, appearance can vary substantially across domains, but geometry-aware intermediate cues help the model retain stable structural understanding, leading to stronger zero-shot segmentation across both satellite and UAV imagery.

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(a)



(b)

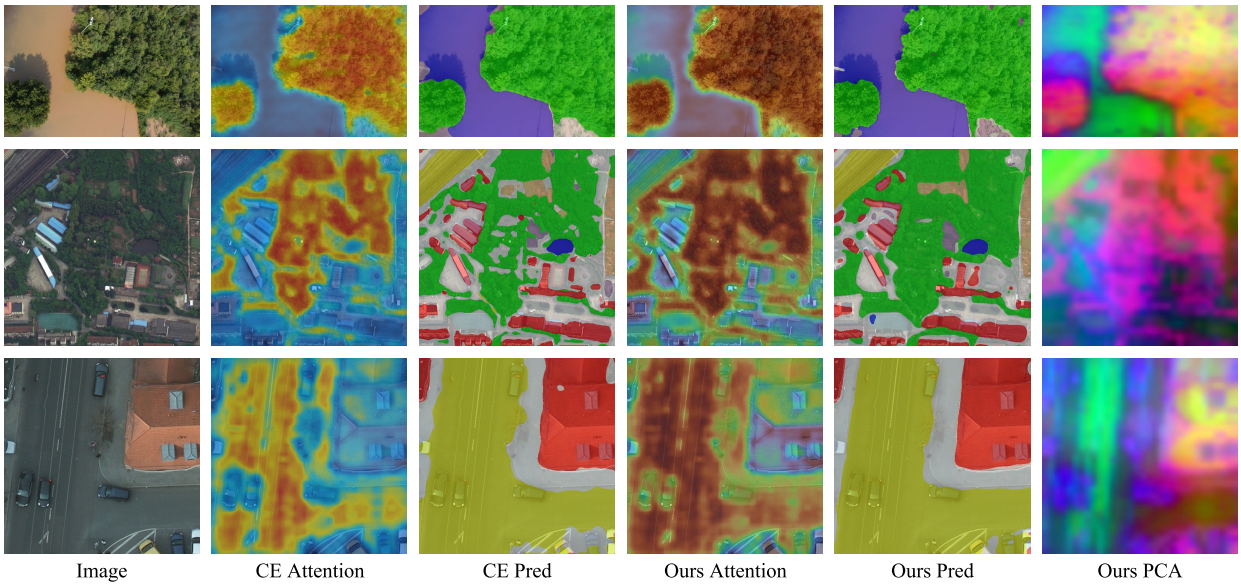
Fig. 5. Representative zero-shot predictions on Guangzhou imagery under two source-domain protocols. (Top) Guangzhou satellite scene evaluated in the LoveDA-source setting, while (bottom) Guangzhou UAV scene evaluated in the VDD-source setting. The zoomed-in-view regions highlight that AerialToken preserves large-scale structural consistency while maintaining local discrimination for buildings, water, roads, and vegetation under substantial cross-view and cross-domain variation. (a) LoveDA-source zero-shot result on a representative Guangzhou satellite scene. (b) VDD-source zero-shot result on a representative Guangzhou UAV scene.

F. Unified Label Space and Cross-Dataset Alignment

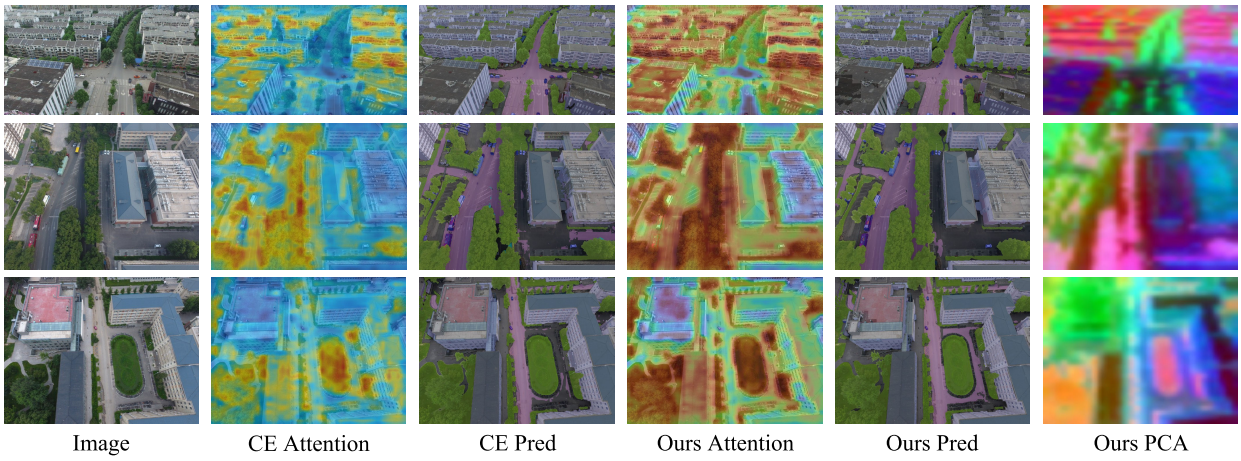
To ensure fair cross-dataset evaluation, we remap each dataset into a shared semantic space under each source protocol. The reported class IDs correspond to the converted evaluation space rather than directly comparable raw IDs across datasets. Categories without a reliable semantic counterpart are mapped to label 0 and excluded from class-wise IoU reporting.

For the LoveDA-source protocol, the unified foreground space contains six classes: building (1), road (2), water (3), barren (4), forest (5), and agricultural (6). Label 0 is reserved for background or ignored categories and is not reported in the class-wise IoU tables. For the VDD-source protocol, we follow the seven-class unified space: other (0), wall (1), road (2), vegetation (3), vehicle (4), roof (5), and water (6). The dataset-specific remapping rules are summarized in Tables XI and XII.

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(a)



(b)

Fig. 6. Attention-based analysis in satellite and UAV settings. Each panel visualizes representative inputs together with baseline responses and AerialToken responses, showing that the proposed geometry-guided features place more stable emphasis on semantically meaningful structures while reducing spurious activations in confusing regions. CE denotes the CrossEarth method. (a) Satellite-view attention and prediction visualization. (b) UAV-view attention and prediction visualization.

The mappings are deterministic label conversions applied once to generate fixed evaluation masks; no pixel-level reannotation or manual relabeling is introduced by this work. All baselines and AerialToken are evaluated on exactly these same converted masks. Ambiguous classes are handled by semantic compatibility rather than by performance-driven choices: impervious surfaces are mapped to road as the closest available hard-surface category in the unified label space, and this mapping is applied identically to all methods, while low vegetation is merged into agricultural only in the LoveDA-source label space where low cultivated or grass-like areas have no separate vegetation category. For the VDD-source protocol, UDD is evaluated with the original UDD6 masks used in our setting, where water is absent, while UAVid uses the VDD authors' IDD-compatible annotations for wall, roof, and water. We only convert class IDs and do not add wall, roof, or water pixels. In particular, the water labels in the UAVid-compatible masks are inherited from the VDD/IDD protocol rather than annotated by us. The three additional unseen domains introduced in this revision follow the same discipline. A candidate benchmark was required to admit a deterministic and semantically faithful mapping into the existing unified label space, and several superficially suitable aerial benchmarks were rejected because they merge buildings and roads into a single built-up class, which would collapse two of the categories the method is designed to separate. Where a source category has no faithful counterpart, it is excluded rather than forced: OpenEarthMap contributes all six LoveDA categories, but its rangeland and developed-space classes are excluded, since LoveDA assigns the equivalent content to its own background class; SkyScapes-Dense and AeroScapes provide no wall or water annotation, so those two categories are absent from their label sets. As with every other target, each mapping is a deterministic class-id conversion applied once; no pixel is reannotated, no target

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label is used at any point in training or model selection, and the same converted masks are used for all compared methods. The evaluated class set therefore differs across targets, and unevaluated categories are marked with a slash in Tables III and V: six classes for FLAIR and OpenEarthMap, five for FloodNet, four for Potsdam and Vaihingen, seven for VDD and UAVid, six for UDD, and five for SkyScapes-Dense and AeroScapes. Each mIoU is the mean over the classes evaluated for that target, so values are comparable across methods within a column but not across targets. Two of the new mappings are deliberately broad: the nine hard-surface categories of SkyScapes-Dense are merged into road, and the Construction class of AeroScapes is mapped to roof, since the unified space has no facade category that a low-altitude view could populate reliably. Both apply identically to every compared method. The SkyScapes impervious surface class spans low wall, rocky terrain, and river, and is left ignored rather than assigned. For classes without a one-to-one counterpart, we either map the raw class to the closest unified category or leave the class absent/ignored, as listed in Tables XI and XII. The deterministic remapping scripts, generated evaluation masks where redistribution is permitted, and evaluation code are released with the code so that the reported protocol can be exactly reproduced.

V. Conclusion

We presented AerialToken, a geometry-aware framework for remote sensing DGSS that transfers intermediate geometry tokens rather than final dense depth maps. By injecting intermediate geometric cues into a frozen VFM and carrying them into geometry-guided decoding, AerialToken improves cross-domain segmentation on both satellite and UAV benchmarks. Experiments on the LoveDA-source and VDD-source protocols show consistent gains over strong recent methods, especially on structure-sensitive land-cover categories such as buildings, roads, and water boundaries. These results suggest that, for overhead remote sensing imagery, calibrated intermediate geometry is a more stable transferable cue than dense-depth predictions where the domain shift alters depth scale. Where acquisition geometry is comparable across domains, the two representations are close, and Table X shows that the measured ordering then depends on which discrepancy statistic is used, so we state the advantage as conditional rather than general.

Three limitations delimit where this design should be expected to help. The geometry route contributes most where categories differ in vertical structure and little where they do not: gains concentrate on buildings, roads and water boundaries and are markedly smaller on low-profile classes such as agricultural land, for which the model falls back on appearance evidence with the same vulnerabilities as appearance-only baselines. The residual injection is bounded relative to the host feature, and that bound must be consistent between training and inference; a model trained with clipping degrades if the bound is removed at test time, while a model trained without it from the start is stable but 2.95 mIoU worse. Finally, the geometry branch adds a second frozen forward pass at inference, which introduces no trainable parameters and no backward cost but is not free at deployment.

Future work will explore open-vocabulary DGSS and source-only geometry adaptation for Earth observation imagery under explicit capacity controls, together with adaptive or learned forms of the screening ratio and the clipping bound.

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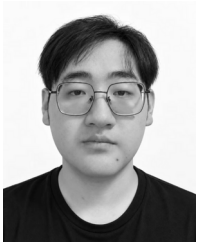
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